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The Stability Gap as a Trajectory Problem: A Bent Descent Path, Its Overshoot, and One Gate in Two Designs

Bachelor's Thesis

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Abstract: Replay-based continual learning is usually judged by the accuracy it reaches *after* each task. Continual evaluation exposes what this view hides: the *stability gap*, a transient drop in past-task accuracy at the start of every new task. The gap persists even when the optimiser descends the exact joint loss, so at least part of its cause lies in the optimisation trajectory rather than the objective. This thesis explains the gap on two levels. The *intended trajectory*, the path exact gradient descent would take, bows out of the old task's basin before returning; and the followed path overshoots the intended one through everything that vanishes with the step size: discrete updates, momentum, and gradient noise. We separate the two levels by measurement rather than by argument. Rescaling the step size is a reparametrisation of the same gradient flow, so a ladder of learning rates run at matched flow time from one frozen checkpoint discretises one and the same intended trajectory, and whatever survives $\eta \to 0$ is that trajectory. On rotated MNIST the ladder converges, and not on zero: of the 23 pp drop at the standard step size, roughly four fifths is discretisation overshoot, and a smooth arc of some four points remains that neither an exact replay gradient nor any further shortening of the step removes. Reaching that limit costs a hundred times the training, which sets the bar for a method: not lowering the gap, which a shorter step does with no method at all, but lowering it at the standard step size and in the standard number of steps. Written in a unified update rule, replay methods counter both levels with a single tool, a *gate* on the new-task gradient with the replay gradient restoring at full strength, and we build and compare the two pure gate designs: an adaptive λ-curriculum and asymmetric Fisher-preconditioned replay. The curriculum earns its keep only under momentum, and there it cuts three quarters of vanilla ER's depth at vanilla accuracy, matches in one epoch what shortening the step reaches only in the limit, and carries to CIFAR-10 on a ResNet-18 at negligible overhead. The directional gate protects the past task at no measured plasticity cost but at far higher compute. Which gate is preferable is set by the momentum regime.

Contents

1 Introduction

Deep neural networks learn powerful representations from static, independent, and identically distributed data, but a vulnerability appears as soon as training becomes sequential: the network overwrites previously acquired knowledge to accommodate new data. This phenomenon, catastrophic forgetting [20, 24], is a major hurdle in machine learning. Retraining from scratch solves the issue, but it is too expensive for systems that need to adapt continuously. Continual learning (CL) studies how to learn from a sequence of tasks without that cost, and its core difficulty is the stability–plasticity dilemma: the model must stay plastic enough to learn new distributions while staying stable enough to preserve old ones [21].

Replay, the family this thesis studies, stores a small buffer of past examples and trains on them alongside the new data. Replay methods reduce long-term forgetting well, but until recently they were evaluated only at task boundaries, which hides what happens within a task. De Lange et al. [5] evaluated the network after every update and identified the *stability gap*: a severe but transient collapse in past-task accuracy that occurs immediately when a new task begins, followed by a slow recovery. This matters in deployment: a system is queried while it trains, so a model caught mid-transition returns wrong answers on inputs its final checkpoint would handle correctly.

The discovery raised a natural question: would a better approximation of the joint loss over all tasks remove the gap? Hess et al. [13] showed that the gap persists even under incremental joint training, where the model has access to all data from all tasks at every step. A perfect objective alone does not remove transient forgetting. The problem therefore lies in *how* the optimisation proceeds, and this thesis locates it on two levels: the *intended trajectory*, the path exact gradient descent would take, already bows out of the old task's basin; and optimisers at a step size greater than zero overshoot the trajectory they intend to follow.

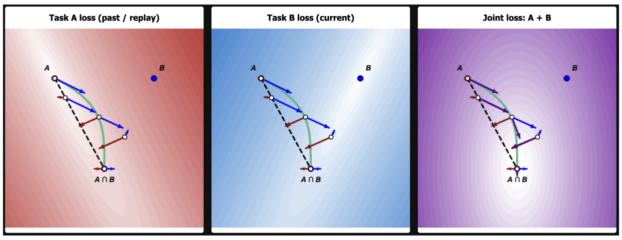


Figure 1.1: 2D optimisation landscape at a task transition. Single-task losses Task A (red) and Task B (blue) are minimised at A and B; the joint loss (purple) is minimised at $A \cap B$. The steepest-descent trajectory (green curve) illustrates the trajectory problem (Section 3.2), bowing outside A's basin into regions of higher loss. In contrast, the dashed black chord sketches the monotone reference path of Section 3.1, which stays inside the basin. Overshoot drivers (Section 3.2) further push the iterate off these paths. An interactive version, in which landscapes, task minima, and both gates of Section 3.3 can be manipulated, is available at `https://arthurreuss.github.io/ValleyWalk/loss_landscape_visualization.html`.

Two levels. Figure 1.1 visualises the transition as a switch from a learned task (Task A) to a new task (Task B), a conceptual reduction of a high-dimensional surface using the loss-landscape lens of Kao et al. [16]. *Level one* is geometric: steepest descent on the abruptly assembled joint loss (green curve) bows out of Task A's basin and returns from outside [16], so the Task-A loss rises above its final level and comes back, its recovery slow and lingering, the *hanging tail.* The excursion is a property of the path, not a necessary cost: Section 3.1 constructs a reference path along which the past-task loss rises monotonically to its joint-optimum level (the dashed chord), valid where its local curvature condition holds, so under that condition everything the accuracy curve does below its settled level and back is avoidable in principle. *Level two* is that optimisers overshoot the trajectory they intend to follow: at a step size greater than zero the large, unopposed boundary gradient [5] is applied in a few discrete jumps rather than traversed as a curve, and finite-buffer gradient noise perturbs every step [1]. Section 3 develops both levels.

One tool: gate the new-task gradient. Sequential replay methods share a single update, a weighted combination of the current-task and replay gradients [26]. Section 3.3 generalises this: because only the ratio of the two weights sets the update direction, any reweighting of this update can be rewritten with the replay gradient restoring at full strength and a single *gate* on the new-task gradient, $d = G\,g_{\text{cur}} + g_{\text{rep}}$. The gate's *domain* can be scalar (attenuate the whole push) or directional (attenuate per direction), and its *control* feedback (react to a live damage signal) or feedforward (act through a schedule or a curvature model). This thesis builds and compares the two pure designs. The first is the **λ-curriculum**, a scalar gate: instead of switching abruptly to the joint loss, it ramps the new-task loss weight $\lambda(t)$ from 0 to 1, so the objective's minimiser moves continuously from θ^*_A to θ^*_{joint} and the model tracks a moving valley floor. Its adaptive variant sets λ from the ratio of the replay and new-task gradient norms, a live damage signal, which makes it a scalar *feedback* gate. The second is **asymmetric preconditioned experience replay** (asymmetric PER), a directional *feedforward* gate: it filters the new-task gradient through the curvature of the replay loss, passing the push at full gain where the past task is flat and braking it where it is steep, leaving the restoring gradient untouched.

2 Background and Related Work

2.1 Continual Learning and Experience Replay

Methods against catastrophic forgetting fall into a few families: replay, parameter and functional regularisation, optimisation-based methods, context-dependent processing, and template-based classification [29]. This thesis focuses on replay, specifically its simplest formulation: Experience Replay (ER) [4]. At each step, ER draws a replay mini-batch from a buffer of stored past examples, combines it with the current-task mini-batch, and takes one optimiser step on the summed loss.

A standard approach to managing this episodic memory is reservoir sampling [30], which ensures that every sample seen so far occupies a buffer slot with equal probability. Under this setup, the buffer remains empty during the first task, meaning ER reduces to plain SGD; consequently, performance gaps and forgetting metrics are evaluated from the second task onward. Despite this simplicity, ER remains highly competitive: a small episodic memory often outperforms purpose-built methods and scales to longer sequences [4]. Furthermore, because the ER update consists of exactly two ingredients, it provides a remarkably clean substrate for isolating and studying interventions at task boundaries.

However, a finite buffer is inherently an imperfect estimator of the past-task distribution. Aljundi et al. [1] frame buffer selection as approximating the feasibility region of the full past data, with reservoir sampling serving as one specific policy within that framework. Consequently, the replay gradient derived from a small buffer will differ in both direction and magnitude from the true gradient of all past data.

2.2 The Stability Gap

De Lange et al. [5] identified the stability gap using continual evaluation, measuring past-task accuracy after every update rather than only at task boundaries. In this view a new task begins with substantial but transient forgetting that the model subsequently recovers from, and they quantify it with worst-case metrics, mainly the minimum past-task accuracy reached during training, rather than with end-of-task averages alone. The effect is not confined to one method: they observe it across experience replay, constraint-based replay, knowledge distillation, and parameter regularisation. It also persists well beyond the standard benchmarks, appearing in joint incremental learning of homogeneous tasks [15] and in continual pre-training of large language models [9], and several mitigations have been proposed [11, 27].

Two findings in this literature anchor the account of Section 3. First, De Lange et al. [5] trace the initial drop to a gradient imbalance at the boundary: the new-task gradient dominates the first updates, so the optimiser trades stability for plasticity precisely when the past task is most exposed. Second, the gap is a property of the trajectory rather than the objective. Hess et al. [13] show it persists under training on the exact joint loss with full access to all past data, and Kao et al. [16] demonstrate the geometric mechanism analytically: steepest descent from an old-task minimum leaves the old basin even with perfect joint gradients, because the steepest direction and the direction to the joint minimum disagree. A perfect objective alone therefore fails to prevent the gap, and the descent geometry itself must be addressed. The optimiser matters as well: comparing optimisers in a similar setting, Obis [23] find the depth and duration of the drop to depend substantially on which one is used, an early indication that the effective step size is a lever on the gap.

2.3 One Update, Many Methods

Beyond plain ER, several methods constrain the update at the task boundary using information about past tasks. GEM stores past examples and, at every step, recomputes each past task's gradient on its stored data, then projects the new-task gradient onto the nearest direction that leaves every stored loss unchanged or lower [18]. A-GEM replaces the per-task constraints with a single averaged past-task gradient, reducing the projection to one cheap operation [3]. EWC adds a quadratic penalty around the past optimum weighted by a curvature estimate [17], and natural-gradient approaches precondition the update with past-task curvature so that steps avoid directions the old task depends on [16].

These methods are not all of one kind: EWC and the natural-gradient family act through a penalty or a preconditioner rather than through replayed data, and they do not fit the form below. The replay-based methods, however, share a single update, as Sodhani et al. [26] show. Every step combines two gradients, the current task's and the replay buffer's, under two scalar weights,

$$\theta \leftarrow \theta - \eta \left(\alpha_1 \nabla L_{\text{current}} + \alpha_2 \nabla L_{\text{replay}}\right), \qquad (2.1)$$

so a replay method reduces to a rule for choosing α_1 and α_2. Plain ER fixes both to one. A-GEM holds $\alpha_1 = 1$ and raises the replay weight α_2 only when the new-task and stored gradients conflict, by just enough to cancel the interference, so its projection is equivalent to re-weighting the two losses; GEM does the same per stored task, through a quadratic program whose multi-constraint solution no longer reduces to two scalar weights. MEGA [10] likewise holds $\alpha_1 = 1$ but sets the replay weight from a loss ratio,

$\alpha_2 = L_{\text{replay}}/L_{\text{current}}$, leaning on memory once the new task is already fit.

The same reading applies to the λ-curriculum of this thesis. It fixes $\alpha_2 \equiv 1$ and turns the other knob, ramping the current-task weight $\alpha_1 = \lambda(t)$ up from zero. It belongs to the same family as A-GEM and MEGA; what distinguishes it is which weight it moves and what signal drives the weight. The adaptive variant (Section 3.3) is the mirror image of MEGA: MEGA raises the *replay* weight from a *loss* ratio to recruit memory once the new task is learned, while the adaptive curriculum holds the *new task* back with a *gradient-norm* ratio, letting it in only as the replay gradient reasserts itself.

2.4 Low-Loss Paths and Mode Connectivity

Mode connectivity gives empirical backing to the *monotone reference path* introduced in Section 3: a route from θ^*_A to θ^*_{joint} along which the past-task loss only ever rises toward its final value requires the two solutions to be joined by a corridor of low loss, and this is what the literature reports. Curved paths of near-constant low loss join independently trained minima [6, 8], straight ones join networks that share early training [7], and, for continual learning specifically, the sequentially trained and multitask solutions often lie in the same basin [22]. Where such a route exists, the stability gap reflects the optimiser wandering off it, and an intervention succeeds by keeping it there.

3 The Stability Gap as a Trajectory Problem

This section reframes the stability gap as a two-level trajectory problem. Learning a new task incurs a permanent stability-plasticity cost; the gap is the transient excess above it. This excess has two sources: a geometric bow in the *intended trajectory*, the path that exact gradient descent would follow, and the overshoot that discrete, finite-step training adds on top of it. We formalise this split by constructing a monotone reference path. We then show that gating the new-task gradient counters these deviations, though no single pure gate design neutralises every driver at once. Instead, how the update is gated dictates which source of error is suppressed and which remains exposed. Two contrasting gates instantiate this: an adaptive scalar curriculum and an asymmetric directional filter. One register note applies throughout: on the quadratic model of this section the claims are exact; carried to deep networks they are hypotheses, which Section 6 tests and Section 7 weighs.

3.1 The Reference Path and the Tracking Bound

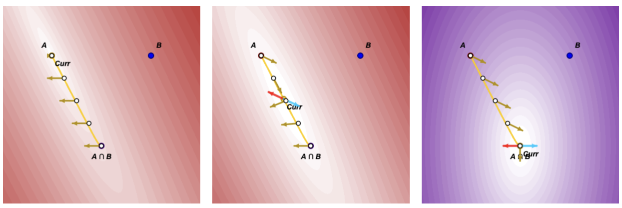
Learning a second task inevitably moves the model. In a quadratic approximation with task losses L_A and L_B and Hessians H_A and H_B, the displacement $d = \theta^*_{\text{joint}} - \theta^*_A$ from the old optimum to the joint minimum permanently raises the replay loss by roughly $\frac{1}{2}\, d^\top H_A\, d$. Every method pays this price; it is distinct from the stability gap.

Yet this rise can be strictly monotone. Consider the curriculum objective $L_\lambda = L_{\text{replay}} + \lambda\, L_{\text{current}}$; at $\lambda = 1$ it is the joint loss with the two tasks weighted equally, the convention used throughout. As λ grows from 0 to 1, the minimiser traces a *reference path* $\Theta^*(\lambda)$ from θ^*_A to θ^*_{joint}. Under local positive-definiteness, an implicit-differentiation argument along the minimiser branch (Appendix D) guarantees that the replay loss climbs directly to its final level with no excursion above it:

$$\frac{d}{d\lambda}\, L_{\text{replay}}(\Theta^*(\lambda)) \geq 0 \qquad \text{for all } \lambda \in [0, 1]. \tag{3.1}$$

This guarantee is local: it holds only while the combined Hessian $H_A + \lambda H_B$ stays positive-definite along the tracked branch of minima. The condition is automatic near θ^*_A and persists until the new task's negative curvature overpowers the old task's; on a deep non-convex backbone it therefore serves as motivation rather than a guarantee. On a ResNet-18 it is almost certainly violated somewhere along the branch, and the group-norm transients of Section 6.4 are where that shows.

Figure 3.1 shows this: as λ grows, the loss contours deform continuously from the old-task basin to the joint basin and the minimiser slides along the connected valley floor. Where the condition holds, the replay loss climbs directly to its final level, and any transient excess *above* it, the stability gap, is a preventable deviation from the reference path.



$\lambda = 0$ $\lambda = 0.025$ $\lambda = 1$

Figure 3.1: Evolution of the curriculum loss $L_\lambda = A + \lambda B$. As λ increases, the contours morph continuously from the old-task basin around A to the joint basin around $A \cap B$, and the current minimiser (*Curr*) slides along the moving valley floor, forming the reference path $\Theta^*(\lambda)$. At the minimiser the replay (red) and current-task (blue) contributions cancel by the first-order condition $\nabla L_{\text{replay}} + \lambda \nabla L_{\text{current}} = 0$. Frames from the interactive visualisation of Figure 1.1.

A real iterate cannot sit on this path exactly, but it can stay close to it. Introducing the new task gradually over an N-step ramp ($\lambda \approx 1/N$) keeps the iterate tethered to the moving valley floor $\Theta^*(\lambda(t))$ rather than letting it leap across the landscape; a gradient-flow analysis bounds the resulting lag by $O(1/N)$ (Appendix D). Because the monotone path fixes the endpoint at the permanent level, this caps the highest transient replay loss of the whole transition,

$$\max_t L_{\text{replay}}(\theta(t)) \;\leq\; L_{\text{replay}}(\theta^*_{\text{joint}}) + O\left(\frac{1}{N}\right), \quad (3.2)$$

the first term the permanent price, the second a transient that shrinks with ramp length. A slow ramp is precisely what the λ-curriculum of this thesis implements.

3.2 The Two-Level Gap: Bowing and Overshoot

Real trajectories deviate from the reference path on two distinct levels, and the step size is what separates them.

Level One (the intended trajectory and its bow): The *intended trajectory* is the path exact gradient descent would follow on the replay objective: gradient flow, the limit of vanishing step size, computed with exact gradients. Even this idealised path bows. The steepest-descent direction of the abruptly assembled joint loss misaligns with the direction toward the joint minimum whenever the loss curves more sharply in some directions than in others, so the path leaves the old task's basin before returning [16]. The gradient imbalance at the task boundary belongs to this level. At θ^*_A the replay gradient vanishes because the iterate sits at the replay optimum, while the new-task gradient is large [5]; the intended trajectory therefore sets off along the new-task direction, and the imbalance shapes where the bow bends. It is a property of the vector field the descent follows, not of how that field is discretised. This deterministic out-and-back arc forms the structural core of the gap (Figure 1.1, green curve).

Level Two (Overshoot of the followed path): The *followed path* is the trajectory the optimiser actually traces. It deviates from the intended trajectory through two fundamental sources of error that only manifest at a finite step size η: discrete updates cut straight chords across the curving landscape (geometric errors that momentum further accumulates), and sampling noise corrupts the buffer-estimated replay gradient (Section 2.1). A single parameter therefore governs this entire level, acting through two distinct drivers. The *acute driver* is the finite step size meeting the massive, unopposed boundary field: the very first update commits to the full new-task gradient over a displacement of $\eta\,\|g_{\text{cur}}\|$, causing the iterate to jump along the new-task direction before the growing replay gradient can pull it back. Within a handful of steps, it sits much further out than the intended trajectory ever goes, producing the sharp first-step drop. The *chronic driver* is the ongoing sampling noise, which corrupts the restoring gradient at every step, slows the recovery tail, and can trigger sudden drops long after the boundary has been crossed.

We hypothesise that these two levels shape the two dimensions of the stability gap: depth records the overshoot, whether from the acute boundary spike or a late, noise-driven drop, while the hanging tail consists of the Level-One arc combined with the slow, noisy recovery the chronic driver leaves behind. This mapping forms our falsifiable core, formalised as RQ1 in Section 4: interventions on the step, slowing it or denoising it, should mitigate the drop but leave the arc intact; placing the optimiser onto the reference path should instead remove the arc; and only combining both should close the gap completely, while the permanent cost of Section 3.1 remains.

3.3 Two Gates: A Division of Labour

Equation (2.1) shows that replay methods differ mainly in how they weight the replay and current-task gradients. Only the ratio of the two weights sets the update *direction*; their common scale sets the length of the step. In gradient flow the scale can be absorbed into the learning rate exactly, so any such method can be rewritten with the replay weight normalised to one. At a step size greater than zero, however, the absorbed scale *is* the effective learning rate, and Section 3.2 identifies the step size as the acute Level-Two driver; the choice of scale is therefore not harmless. We keep the normalisation but fix the learning rate across all methods and conditions, so every comparison runs at a matched step size. Furthermore, we measure the step size's own contribution in a dedicated block, the step-size ladder of Section 5, which drives η towards zero at matched flow time and leaves the objective, the buffer, and the update direction untouched [23].

The replay gradient is the only restoring force pulling the iterate back to the reference path, so a protective method must attenuate the new-task push, the source of the bow and overshoot, while leaving the restoring gradient intact. Generalising the current-task weight to an operator G yields a one-sided *gate*:

$$d \;=\; G\,g_{\text{cur}} \;+\; g_{\text{rep}}. \quad (3.3)$$

The curriculum of Section 3.1 realises this gate along one axis, *time*: it introduces the new task slowly, so the push is never large enough at once to knock the iterate off the reference path. A complementary axis is *direction*. At any single step only part of the push is harmful, because the past task is flat in most directions and steeply curved in a few, and only motion along the steep directions raises the replay loss. Preconditioning the push by past-task curvature exploits this, passing it at full gain where the past task is flat and shrinking it where the past task is steep, so retention costs little plasticity. This natural-gradient idea [14] underlies Natural Continual Learning [16] and, in a replay setting, Online Curvature-Aware Replay [28]. Both, however, precondition the *entire* update, so the same metric that brakes the harmful push also brakes the restoring replay gradient, protecting the past task only by slowing the return to it. Asymmetric PER keeps the gate one-sided: it filters the new-task gradient alone and leaves the restoring gradient at full strength. We include the symmetric preconditioner, which damps both, as a control (Appendix C.5).

This thesis explores two specific gating mechanisms to implement this attenuation, alongside a fixed linear ramp that serves as an ablation:

Adaptive λ-curriculum: A scheduling parameter $\lambda(t)$ sets the new-task weight from the ratio of replay to new-task gradient norms, a live damage signal that fires whichever direction harm arrives from. By scaling the loss itself, it deforms the objective into the homotopy of Section 3.1, so the optimiser's natural descent follows the reference path.

Asymmetric PER: The gate $G = \delta(F_{\text{rep}} + \delta I)^{-1}$ is a filter set by the replay Fisher F_{rep}, giving the update

$$d \;=\; \delta\,(F_{\text{rep}} + \delta I)^{-1} g_{\text{cur}} \;+\; g_{\text{rep}}, \tag{3.4}$$

with the damping δ its single knob: as $\delta \to 0$ the push concentrates in the directions the past task is flat in. The filter is rebuilt each step from the current parameters and past-task curvature.

These distinct mechanisms yield specific hypotheses regarding how each gate navigates the stability-plasticity dilemma, formalised as research questions in Section 4. While both gates aim to steer the intended trajectory back toward the monotone reference path, they tackle the Level Two deviations from fundamentally different angles. Because the curriculum holds back the entire new-task push at the source, it successfully mitigates the acute boundary overshoot, but it does so at a direct plasticity cost. Furthermore, it leaves chronic sampling noise exposed, which can lead to late, noise-driven drops. Consequently, it requires a separate accelerator to repay the plasticity debt (RQ2). Conversely, the asymmetric filter isolates the harmful push by direction, theoretically managing the noise and protecting the past task with a minimal plasticity penalty (RQ3). However, this step-level filtering leaves the method prone to acute overshoot at the start. This theoretical elegance also trades off against computational cost: while the curriculum requires only a cheap gradient-norm ratio per step, asymmetric PER must solve a linear system in the replay Fisher via conjugate gradients (Appendix A.5).

Crucially, we hypothesise that optimiser momentum will expose a failure axis that discriminates between the two gates based on *where* they attenuate the gradient. Because the curriculum attenuates at the *source* ($\lambda \approx 0$ early in training), the new-task gradient is structurally flattened at the loss level. With no large gradient to accumulate, momentum can safely compose with the gate. Here, momentum solves both of the curriculum's weaknesses simultaneously: it provides acceleration to repay the plasticity debt, and it acts as a smoothing mechanism to average out the chronic gradient noise, all without re-inflating the acute overshoot. In contrast, asymmetric PER attenuates at the *step*. The raw new-task gradient remains large, and the filter shrinks it step-by-step. Because momentum accumulates past velocities, it ultimately rebuilds the harmful displacement that the per-step filter attempts to remove. For the directional gate, momentum is therefore expected to restore plasticity while actively worsening the acute overshoot, undermining the initial protection and deepening the gap.

This division of labour illustrates that mitigating the stability gap is about understanding the specific mechanisms at play. These two approaches reveal how their specific gating mechanisms interact with the momentum regime.

4 Research Questions

The two-level account and tracking bound of Section 3 make falsifiable predictions about the stability gap and the behaviour of distinct gating mechanisms. To evaluate these predictions, our experiments are organised around four core questions. The first three are evaluated in a controlled setting (rotated MNIST with a small MLP) to probe the gap's dynamics step-by-step, while the fourth tests the generalisability of these findings under increased complexity.

RQ1: The account. A uniform rescaling of the learning rate is a time reparametrisation of the same gradient flow, so driving $\eta \to 0$ at matched flow time follows the intended trajectory ever more faithfully without changing it. **Does the stability gap separate under that limit into a deterministic arc that survives it, belonging to the intended trajectory, and an overshoot that exists only at finite step size?**

RQ2: The adaptive scalar gate. Taking the limit is itself a repair, but a costly one: a smaller step removes the overshoot and bills for it in training length, the epoch count scaling as $1/\eta$ at matched flow time. **Can an adaptive scalar feedback gate (the adaptive λ-curriculum) fix the trajectory without paying that bill, and how does the answer depend on optimiser momentum and on the plasticity the gate withholds?**

RQ3: The directional feedforward gate. **To what extent does a directional feedforward gate (asymmetric PER) reduce the stability gap, and how does accumulated momentum compromise this gating mechanism?**

RQ4: Generalisation. **Do the mechanisms underlying the stability gap, and the effectiveness of the proposed gates, generalise to deeper network architectures and more severe task shifts?**

5 Methods

This section describes the empirical testbeds (Section 5.1), the metrics that summarise each run (Section 5.1.2), and the experimental blocks (Section 5.2) that answer the research questions of Section 4. All code, experiment configurations, and analysis scripts are available at `https://github.com/Arthurreuss/ValleyWalk`.

5.1 Experimental Setup

5.1.1 Testbeds

Two testbeds of different scale carry the study. A small, fast one (rotated MNIST on a two-layer MLP) runs the dense mechanistic sweeps: the step-size ladder, the curriculum sweep, the asymmetric-PER damping sweep, and the momentum cross. A larger one (domain-shift CIFAR-10 on a CIFAR-adapted ResNet-18) tests whether the findings survive a deeper backbone and a stronger task shift, the regime where the local condition behind the bound of Section 3.1 weakens. Both hold the label space and task structure fixed and shift only the input distribution, and both use two tasks in their main blocks, so there is exactly one transition and one instance of the stability gap; a five-task rot-MNIST sequence and a three-task CIFAR sequence serve as multi-boundary checks.

On rot-MNIST, T_0 presents the digits upright and T_1 rotates them by $90°$, each task using the standard 60,000 training and 10,000 test images; the $90°$ domain shift follows De Lange et al. [5], which orients the pair as $-90° \to 0°$. On CIFAR-10, T_0 is clean and T_1 applies Gaussian-noise corruption at severity 3 [12]; the ten classes are shared, so only the input distribution shifts. A three-task variant ($[\mathrm{none}, \mathrm{Gaussian}, \mathrm{shot}]$) and two alternative corruptions appear in the generalisation block. The rot-MNIST MLP is cheap enough to sweep across many conditions and seeds and is where the local condition of Section 3.1 is most plausible; the ResNet-18 is the deeper, more realistic counterpart. rot-MNIST runs online (one epoch per task); the ResNet needs several passes to converge before the transition, so the CIFAR block uses ten epochs per task. Optimiser, learning rate, and batch size are common across benchmarks; no method retunes the learning rate, so every comparison between methods runs at a matched step size (Section 3.3). The step size is held fixed in that role and manipulated in one block only, the ladder of Section 5.2.1, which holds the objective, the buffer, and the update rule fixed and varies η alone. Full backbone and protocol specifications are in Appendix A.2 (Tables A.1–A.3).

Throughout training every task's test accuracy is evaluated every 10 steps; at each transition the cadence rises to every step, so the dip is recorded at full resolution (the whole online epoch on rot-MNIST, a 250-step window on CIFAR). A stability-gap tracker snapshots each past task's pre-switch accuracy and records the resulting (step, accuracy) series; how the per-step figures draw the sparsely sampled pre-boundary segment is a plotting convention with no effect on any reported number, stated in Appendix A.2. On the vanilla-ER reference runs only, a gradient tracker additionally records the fidelity of the replay-buffer gradient against the exact past-task gradient, at the same cadence; each such step costs a full pass over the past-task training set, which is why it is not run elsewhere, the ladder's long rungs least of all.

5.1.2 Metrics

Each metric is reported as the seed mean ± sample standard deviation; complete definitions are in Appendix B. The gap definition of Section 3, the transient excess below the level a task settles at, gives two primary metrics. With a_j^{pre} the pre-switch baseline on past task j and $a_j(t)$ its accuracy at step t, `gap_depth` is the maximum instantaneous excursion $\max_j(a_j^{\mathrm{pre}} - \min_t a_j(t))$, which Section 3.2 predicts to be overshoot-dominated. The second, `gap_area_end`, integrates the shortfall below the settled level: with reference $b_j = \min(a_j^{\mathrm{pre}}, \bar{a}_j)$, where $\bar{a}_j$ is the accuracy the task eventually reaches, it sums $\max(0, b_j - a_j(t))$ over the new task's training steps and across past tasks. Anchoring to the recovered level implements the split of Section 3.1: a permanent step down to a lower plateau contributes to the standard forgetting metrics and near-zero to the area, while a true dip-and-recover registers in full. One consequence, revisited when reading the tables, is that two methods settling at different levels are measured against different lines, so a better-recovering method can carry the *larger* area; a positive area marks transient disruption, not net cost, and we read it with final accuracy. The arc of Section 3.2 and the slow part of the noisy recovery live here; sharp noise-driven drops register in depth instead. A third metric, `recovery_steps`, and the gradient-fidelity diagnostics are defined in Appendix B.

One clarification is needed for the ladder of Section 5.2.1, the only block in which η itself varies. `gap_depth` is a maximum over accuracies and is therefore η-invariant by construction. `gap_area_end` is not: it is a trapezoidal integral of the drop curve against the step index, weighted by the spacing between records, with units accuracy·steps. Stretching a rung by $c = 0.1/\eta$ stretches its step axis by the same factor, so an identically shaped excursion reads c times larger. Across the ladder the area is therefore taken in flow time, $A_\tau = \eta\, A_{\text{steps}}$, the integral of the same curve against $\tau = \sum_t \eta$; within a rung the two differ by a constant, and every block at fixed η reports the step-axis value as before.

To place the gap in the standard frame we report the continual-evaluation metrics of De Lange et al. [5] from the task-boundary accuracy matrix: average final accuracy (ACC), which checks that a gate lowers the gap at preserved accuracy; worst-case retention (min-ACC, WC-ACC); and short-horizon plasticity (WP_{100}). FORG and the windowed WF_w/WP_w family are logged for completeness. Conditions are compared with the paired Wilcoxon signed-rank test on seed-paired differences, preferred over the t-test because five paired seeds cannot support a normality assumption on bounded metrics: the gap metrics floor at zero, where the strongest conditions cluster, and accuracy sits near its ceiling, so the seed differences are skewed rather than Gaussian. With five paired seeds the smallest two-sided p-value is 0.0625, reached when all five differences share a sign; we read $p = 0.0625$ as complete directional agreement and let effect sizes separate large results from marginal ones.

5.2 Experiments

Each block maps to one or more research questions and tests the corresponding predictions of Section 3. Unless otherwise noted, all conditions use five seeds and follow the standard protocol. Vanilla ER at the standard operating point (1k reservoir, $\eta = 0.1$, no schedule, and no filter) serves as the common reference baseline for every rot-MNIST block and is run at both momentum settings.

5.2.1 The Step-Size Ladder (RQ1)

To address RQ1, the step-size ladder isolates the effects of discretisation from the intended optimisation trajectory. A uniform rescaling of η does not change the underlying vector field. Because forward Euler at step η integrates the same flow $\dot{\theta} = -g(\theta)$ on a finer grid, a smaller step size follows the intended path more accurately. Therefore, as η decreases, any reduction in the performance gap represents discretisation error. Whatever error survives at the limit belongs to the continuous path the optimiser was trying to follow, which we define as the arc.

To ensure the rungs of the ladder represent a true limit rather than unrelated runs, we enforce four constraints. First, we match the total flow time: the epoch count scales as $1/\eta$, meaning a rung at $\eta = 0.1/c$ runs for c epochs. This guarantees every cell covers the same flow time $\tau = 23.5$ over the new task, preventing artificial performance gaps caused simply by under-training. Second, we use a frozen θ_0^*: every cell warm-starts from a shared task-0 checkpoint (Appendix A.4) rather than training the initial task at its own step size. This prevents smaller step sizes from settling into differently sharp minima before the task switch. Third, we maintain a matched record grid, scaling the evaluation cadence by c so every cell records the exact same number of samples spaced equally in flow time. Finally, the evaluation cadence is counted directly from the task boundary to ensure we capture the initial post-switch spike perfectly.

Table 5.1 details the ladder rungs: $\eta = 0.1$, $\eta \approx 0.033$, $\eta = 0.01$, and $\eta = 0.001$. Every rung runs at both momentum settings, the finer rungs establishing the limit each leg converges on. Each rung is evaluated on two buffer arms (the standard 1k reservoir and a 60k exact past-task gradient buffer). Using the exact past-task gradient switches off estimator variance at the source, ensuring that any excursion surviving both the small step size and the exact gradient is purely a property of the path.

The momentum leg provides a secondary perspective. Because momentum multiplies the asymptotic step by $1/(1-\mu)$, the $\mu = 0.9$ rungs operate at effective steps of 1.0, 0.33, 0.1, and 0.01. This allows us to compare exact effective-step diagonals between the two momentum settings. A momentum cell that behaves like its matched $\mu = 0$ partner indicates that momentum's primary effect is bounded by the effective step size, whereas deviations point to the velocity accumulator itself.

Since forward Euler's global error is $O(\eta)$, we estimate the limit by fitting a linear residual metric $A(\eta) = A_0 + k\eta$ over the small-η rungs. A departure from linearity at $\eta = 0.1$ signifies the discrete instability responsible for the first-step spike. The two levels predict opposite behaviors: depth (the overshoot) should fall toward zero down the ladder, while the area in flow time (the arc) should converge on a positive A_0 that no further step-size reduction can eliminate.

5.2.2 The λ-Curriculum Sweep (RQ2)

The curriculum sweep evaluates the scalar feedback gate by testing its components individually (Table 5.2). The central claim of the curriculum is not just that it lowers the performance gap, but that it successfully does so at the standard step size, within the standard step budget, and by modifying only the new-task update. All conditions run on top of standard ER, making the schedule the only independent variable.

Conditions C1 through C3 test linear ramps to evaluate the $O(1/N)$ prediction of Equation (3.2). Condition C4 introduces the adaptive schedule (the feedback variant of Section 3.3), which sets λ from a clipped exponential moving average of $\|g_{\text{replay}}\|/\|g_{\text{new}}\|$, reset to zero at each boundary. Because $\|g_{\text{replay}}\| \approx 0$ at θ_0^*, it starts near zero; unlike the ramps it has neither a length nor a terminal state, tracking the ratio for as long as the task runs and falling again if the ratio does. How far it releases the new task is therefore an outcome of the run rather than a setting, and the two momentum legs differ sharply in it (Appendix A.3). Conditions C5 through C7 apply a minimum floor ($\lambda_{\min}$) to preserve early plasticity. Condition C8 serves as a diagnostic substitution argument, pairing the optimal practical configuration (C7) with the exact 60k replay gradient. If the curriculum leaves a residual transient at momentum 0, applying the exact gradient tests whether that residual is merely buffer noise. Comparing this to momentum runs helps isolate momentum's variance-reduction properties from its role in accelerating new-task learning.

Label	Condition
Baseline	Vanilla ER ($\lambda \equiv 1$)
C1	Linear N=50
C2	Linear N=100
C3	Linear N=200
C4	Adaptive
C5	Adaptive + $\lambda_{\min}$=0.05
C6	Adaptive + $\lambda_{\min}$=0.10
C7	Adaptive + $\lambda_{\min}$=0.20
C8	C7 + full 60k buffer

Table 5.2: λ-curriculum conditions, all applied on top of standard ER and all run at momentum 0.0 and 0.9. The linear conditions ramp λ as $\text{clip}(t/N, 0, 1)$; the adaptive ones use $\max(\lambda_{\min}, \text{clip}(\text{EMA}_\alpha(r), 0, 1))$ with $\alpha = 0.05$ and $r := \|g_{\text{replay}}\|/\|g_{\text{new}}\|$. C4–C7 settle the two ingredient choices of the shipped configuration and are reported in Appendix C.2.

5.2.3 Momentum Cross (RQ2, RQ3)

To test the composability prediction of Section 3.3, we run the vanilla ER reference, the step-size ladder, the curriculum sweep, and the asymmetric-PER sweep at both momentum 0.0 and 0.9 (denoted with an M subscript). This cross-examination reveals whether momentum helps or hinders a method based on where that method attenuates the parameter push. We also conduct a multi-boundary check using a five-task rot-MNIST sequence to repeat this three-way comparison across multiple transitions (Appendix C.9).

5.2.4 Asymmetric-PER δ Sweep (RQ3)

The damping sweep (Table 5.3) evaluates the directional feedforward gate by testing how the gap shrinks as the filter strengthens and identifying where the gate ultimately fails. Asymmetric PER implements Equation (3.4), where the gain along an eigendirection of replay curvature ρ is defined as $\delta/(\rho+\delta)$. This means $\delta \to \infty$ recovers plain ER, while $\delta \to 0$ forces the update into replay-flat directions while keeping the restoring gradient at full strength.

Label	η	Epochs c	Cadence	Steps	τ
S1	0.1	1	1	235	23.5
S2	0.1/3	3	3	705	23.5
S3	0.01	10	10	2350	23.5
S4	0.001	100	100	23500	23.5

Table 5.1: The step-size ladder on two-task rot-MNIST. Cadence is the dense post-switch evaluation interval in steps, so every rung records the same 235 samples over T_1. Each rung runs on both buffer arms at both momentum settings: sixteen cells, five seeds, warm-started from one frozen θ_0^* per momentum setting and seed.

Both legs sweep $\delta \in \{1.0, 0.3, 0.1, 0.03\}$, from a filter that barely moves the update to the strongest setting reached here. The standard leg matches the practical memory budget of the curriculum, while the full-buffer leg uses the exact past-task gradient to eliminate variance, allowing us to measure the deterministic arc directly. We include two controls at the settings where the effects they check for would appear: a solver control rerunning the strongest full-buffer cell ($\delta = 0.03$) with 60 conjugate-gradient iterations instead of 25, to rule out numerical artefacts, and a symmetric control over the same four δ on the standard leg that preconditions the summed gradient, isolating the specific effect of the one-sided gate. When momentum is active, the filtered direction enters the velocity accumulator, meaning momentum acts downstream of the gate.

Label	Condition	Replay buffer
P1	Standard leg	1k reservoir
P2	Full-buffer leg	60k (exact g_{rep})
P3	Solver control	60k
P4	Symmetric control	1k reservoir

Table 5.3: Asymmetric-PER conditions. Both sweep legs run at momentum 0.0; momentum-0.9 counterparts cover $\delta \leq 0.3$ on the standard leg and $\delta \in \{0.3, 0.1\}$ on the full-buffer leg. The two controls run at momentum 0, where the effects they check for arise. The filter metric is always the replay Fisher F_{rep}, estimated on a sampled batch (Appendix A.5).

5.2.5 Generalisation: CIFAR-10 (RQ4)

The CIFAR-10 block tests whether the gating mechanisms and momentum dynamics scale to a deeper backbone and a more severe task shift (Table 5.4).

We evaluate each method using both BatchNorm and GroupNorm to isolate optimisation dynamics from normalisation transients. Because the input shift disrupts BatchNorm's running statistics, it forces a readaptation phase during the first few steps. GroupNorm avoids cross-batch statistics entirely, ensuring that its transient reflects pure optimisation behavior.

The curriculum and asymmetric PER carry their tuned rot-MNIST configurations directly into this block without CIFAR-specific hyperparameter sweeps, ensuring a rigorous test of their generalisation. Furthermore, we test the practical curriculum against vanilla ER on novel corruptions (such as a shot-noise shift and a contrast shift) to confirm the results are robust across different types of distribution shifts rather than overfit to Gaussian noise.

Label	Method	Norm	μ
G1	Vanilla ER	BatchNorm	0.9
G2	Adaptive $\lambda_{\min}$=0.20	BatchNorm	0.9
G3	Asym. PER δ=0.1	BatchNorm	0.9
G4	Vanilla ER	GroupNorm	0.9
G5	Adaptive $\lambda_{\min}$=0.20	GroupNorm	0.9
G6	Asym. PER δ=0.1	GroupNorm	0.9
G7	Vanilla ER	GroupNorm	0.0
G8	Adaptive $\lambda_{\min}$=0.20	GroupNorm	0.0
G9	Asym. PER δ=0.1	GroupNorm	0.0

Table 5.4: CIFAR-10 conditions (two-task Gaussian-noise shift, buffer 5,000, five seeds). Top: the headline block at momentum 0.9. Bottom: the momentum cross, pairing momentum-0 runs of all three methods against their momentum-0.9 counterparts at GroupNorm. The generalisation block (novel corruptions and the three-task sequence, three seeds) is described in the text and reported in Appendix C.7.

6 Results

The results section addresses our core research questions by breaking down the forgetting transient into its fundamental components (Section 6.1, RQ1), evaluating the scalar feedback gate (Section 6.2, RQ2) and the directional feedforward gate (Section 6.3, RQ3), and confirming that both methods scale to a ResNet-18 on CIFAR-10 (Section 6.4, RQ4).

Reading the metrics. To focus on the underlying phenomena, this section prioritises narrative insights over exhaustive statistical breakdowns (full metric tables are available in Appendix C). Gap depth refers to the worst past-task performance drop following a task switch. Area integrates the excursion over time, capturing how long a task remains compromised.

6.1 The Two Levels, Separated (RQ1)

RQ1 investigates the underlying mechanisms of the performance drop during a task switch, specifically disentangling the fundamental properties of the trajectory from overshoots introduced by discrete optimization steps. We answer this by shrinking the learning rate η toward zero and plotting the trajectory against matched optimization flow time rather than step count, the axis on which rungs that differ in length are directly comparable (Appendix A.2).

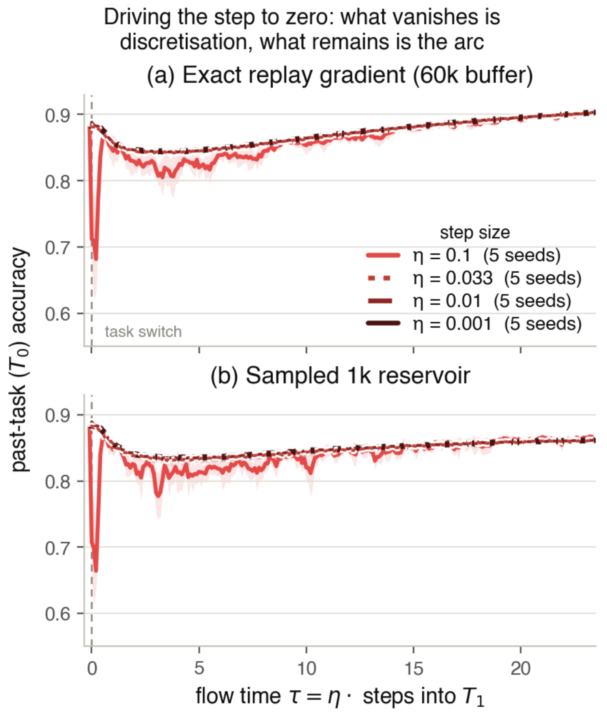


Figure 6.1: The effect of step size (η) on past-task accuracy during a task switch. Across both (a) exact gradients and (b) standard reservoir replay, reducing the step size eliminates the sharp initial boundary spike caused by the standard learning rate (η = 0.1). Crucially, a smooth performance dip (the "arc") remains across all finer step sizes, demonstrating that this underlying drop is a fundamental property of the trajectory, rather than an optimization artifact.

The ladder converges, and not on zero. To understand what drives the gap, we must distinguish between two metrics: the *depth* of the performance drop and the *area* of the excursion. Figure 6.1 illustrates this distinction perfectly. When we reduce the step size from the standard 0.1 point to much finer resolutions, the massive initial spike vanishes. This dramatic reduction in depth proves that the bulk of standard catastrophic forgetting is a discrete instability, a direct result of a large step size hitting an unbalanced boundary gradient.

However, as the ladder of step sizes converges, it does not converge on zero. Even as the step size vanishes, a smooth, shallow dip remains. This is easiest to see in Figure 6.1(a), where we use an exact replay gradient to eliminate any variance from memory sampling. Even without estimator noise, and with a near-zero step size, the trajectory still registers a positive limit in both its residual depth and its flow-time area. The step size cures the discrete overshoot, but this stubborn, unremovable excursion is the inherent "arc" of the trajectory.

Accuracy is not what the limit costs. Crucially, shedding the discrete instability to reach this minimal arc does not sacrifice final accuracy, the fine-grained runs recover just as well as the standard run. Instead, the cost is purely computational: traversing the exact same trajectory with tiny steps requires drastically more epochs. Lowering the gap is therefore not by itself an achievement, since a shorter step does it with no method at all. Following RQ1, the criteria for our gating mechanisms (RQ2 and RQ3) are to simultaneously resolve the discrete overshoot and mitigate the underlying arc, achieving both under a practical step-size.

6.2 The Scalar Feedback Gate (RQ2)

The scalar feedback gate (adaptive curriculum) successfully protects past knowledge by scaling down the new-task loss term when boundary gradients threaten stability. Table 6.1 puts its practical configuration (C7) beside the directional gate of Section 6.3 and the ungated baseline, at both momentum settings; the ingredient sweep behind C7 is in Appendix C.2.

The schedule works, but incurs a plasticity debt without momentum. As shown in Figure 6.2 (dashed lines), testing the curriculum without momentum reveals the core trade-off. By scaling down the new-task loss term during the boundary transition, the gate effectively shrinks the *depth* of the initial forgetting spike and reduces the total *area* of the excursion compared to the vanilla baseline. However, Figure 6.2(b) illustrates the cost: early learning on the new task is visibly delayed, creating a plasticity debt that depresses final accuracy. Furthermore, while the curriculum prevents the catastrophic initial drop, the dashed blue line in Figure 6.2(a) shows it still suffers from a jittery, chronic baseline of forgetting. This residual is an artifact of estimator variance, not a secondary deformation of the path: the curriculum scales the new-task term but leaves the replay estimate as noisy as the vanilla baseline. A control experiment using exact replay gradients verifies that this variance accounts for the residual depth (see Appendix C.2 for full details).

Momentum pays the debt and closes the gap. The scalar gate is explicitly designed for a momentum-enabled regime, where it excels by solving both the variance and plasticity problems simultaneously. As seen in the solid lines of Figure 6.2, momentum acts in two complementary roles. First, it averages out the noisy replay estimates over time, collapsing the remaining jittery variance and holding the past task nearly flat against its plateau (Figure 6.2(a)). This drives both the *depth* and *area* of the gap down to a mere fraction of the vanilla baseline. Second, momentum accelerates the learning process, entirely repaying the plasticity debt incurred by the curriculum's early braking (Figure 6.2(b)). Ultimately, the combination of the adaptive curriculum and momentum achieves profound stability while fully restoring final accuracy to vanilla levels as seen in Table 6.1.

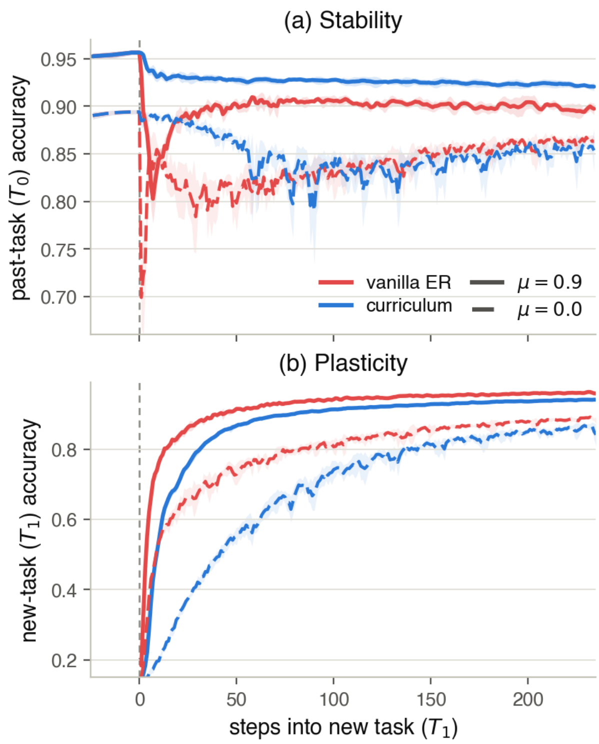


Figure 6.2: The scalar feedback gate and momentum's two roles: per-step accuracy across the rot-MNIST switch, seed mean with ±std bands, for vanilla ER (red) and the practical adaptive curriculum $\lambda_{\min}$=0.20 (blue), each at momentum 0.9 (solid) and 0.0 (dashed). (a) Stability (T_0): only curriculum-with-momentum (solid blue) holds the past task near its plateau through the switch; momentum leaves vanilla's spike, and both methods open a deep, jittery gap without it (dashed). (b) Plasticity (T_1): at momentum 0 the curriculum slows early new-task learning (dashed blue), the plasticity debt that momentum pays back (solid blue).

Condition	depth (pp)	area	ACC	WP_{100}
Momentum 0.0				
Vanilla ER (no gate)	19.5 ± 4.1	6.27 ± 0.47	0.873 ± 0.005	0.736 ± 0.020
C7: curriculum, $\lambda_{\min}$=0.20	13.9 ± 3.4	2.71 ± 0.57	0.837 ± 0.031	0.710 ± 0.010
P1: asym. PER, δ=0.03	$\mathbf{4.8 \pm 1.9}$	$\mathbf{1.04 \pm 0.58}$	$\mathbf{0.880 \pm 0.005}$	$\mathbf{0.741 \pm 0.007}$
Momentum 0.9				
Vanilla ER (no gate)	15.9 ± 2.6	1.03 ± 0.15	0.928 ± 0.003	0.808 ± 0.012
C7: curriculum, $\lambda_{\min}$=0.20	$\mathbf{3.7 \pm 0.4}$	$\mathbf{0.03 \pm 0.02}$	$\mathbf{0.931 \pm 0.002}$	0.812 ± 0.009
P1: asym. PER, δ=0.03	8.5 ± 0.6	0.23 ± 0.12	0.929 ± 0.003	$\mathbf{0.816 \pm 0.010}$

Table 6.1: The two gates against the ungated baseline on rot-MNIST, at momentum 0.0 and 0.9; five seeds, standard (1k) buffer, best in each block in bold. Each gate is shown at the configuration its own section settles on: the practical curriculum (C7) and the strongest filter (P1). Without momentum, the directional gate clearly outperforms the scalar variant, uniquely providing stability without incurring a trade-off in overall accuracy or WP_{100}. Under momentum, however, their effectiveness inverts. The scalar gate emerges as the superior method for reducing the drop while preserving baseline accuracy, while the directional gate loses much of its protective capacity.

6.3 The Directional Feedforward Gate (RQ3)

The directional feedforward gate (asymmetric PER) employs a targeted regularization strategy. Rather than attenuating the new-task push uniformly at its source, as the scalar curriculum does, it selectively filters gradient components that align with the high-curvature directions of the past task. Table 6.1 presents the results for the most stringent filter setting evaluated using the standard experience replay buffer, matching the baseline operating conditions. The complete hyperparameter sweep across both momentum settings is detailed in Appendix C.4.

Protection without plasticity degradation. Figure 6.3 contrasts the strongest filter configuration against the ungated baseline, structured to facilitate direct comparison with the scalar gate. In the absence of momentum (dashed line), the filter maintains past-task performance notably above the vanilla ER baseline throughout the transition phase. Furthermore, Figure 6.3(b) demonstrates that this stability does not incur a long-term penalty: while new-task learning experiences a brief initial lag, it quickly converges with the baseline trajectory. Because this deceleration is restricted to directions flagged by past-task curvature rather than the global loss landscape, it fully resolves before the WP_{100} horizon. This behavior represents an inversion of the plasticity debt observed with the scalar gate. Consequently, as filter strength increases, performance drop depth and area decrease while overall accuracy and WP_{100} consistently improve (Table C.5). Consequently, the gate approximates the idealized "arc" trajectory without relying on microscopic step sizes or sacrificing plasticity. The monotonic relationship governing this behavior is shown in Figure C.4 (see Appendices C.4 and C.5 for supplementary symmetric, higher conjugate gradients, and exact gradient controls).

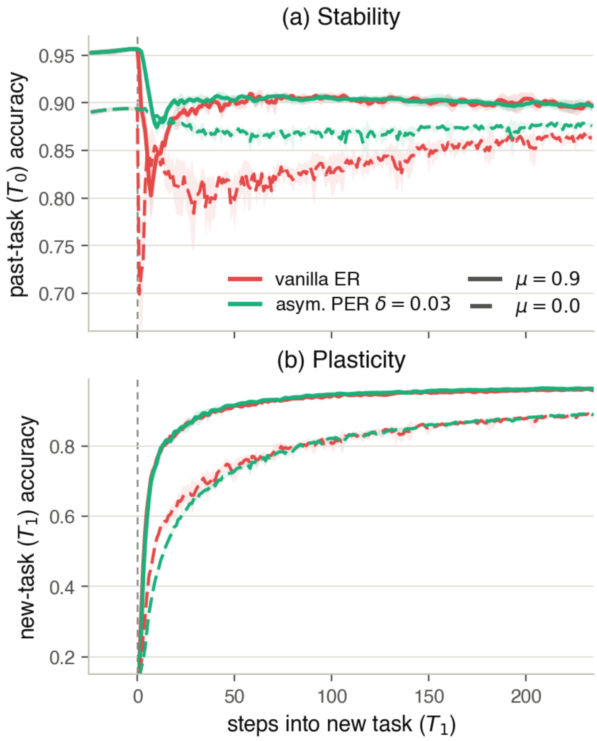


Figure 6.3: Interaction between the directional feedforward gate (asymmetric PER) and momentum across a task switch. (a) Stability (T_0): Without momentum, the strongest filter effectively mitigates both the initial performance spike and the subsequent chronic dip observed in vanilla ER. With momentum enabled, this protective effect is severely diminished, as the momentum accumulator gradually reintroduces the displacement that the per-step filter attempts to remove. (b) Plasticity (T_1): Unlike the global deceleration caused by the scalar gate (Figure 6.2(b)), the directional filter induces only a brief, selective initial lag on new-task learning before matching the vanilla baseline trajectory, regardless of the momentum setting.

Momentum counteracts directional filtering. The protective benefits of the directional gate are largely negated when momentum is applied (full line). Replicating the evaluation with momentum enabled reveals that the steep initial overshoot is reintroduced. This failure stems from the localized, per-step nature of the gating mechanism: while the filter successfully restricts the gradient update at any single iteration, the momentum accumulator integrates these moderated updates over time, effectively reconstructing the rapid initial displacement it was designed to prevent. Although the gate still offers a relative reduction compared to the vanilla ER baseline, its protective effect is greatly reduced, and it offers no improvement in overall accuracy. While momentum causes the deep initial drop to return, it also structurally compresses the transient phase. Consequently, the integrated area remains small: the model recovers more rapidly than vanilla ER and completely avoids the chronic underlying arc.

6.4 Generalisation to CIFAR-10 (RQ4)

To ensure these findings are not artifacts of simple environments, we tested both gates on a ResNet-18 architecture transitioning across CIFAR-10 corruptions. Table 6.2 and Figure 6.4 confirm that the mechanisms carry to deeper networks.

Configuration of the directional gate. It should be noted that the directional gate is evaluated here at a slightly weaker filter setting (δ=0.1) than the optimal configuration (δ=0.03) established in Section 6.3. However, because the performance of the gate scales monotonically with filter strength in this regime (as detailed in Appendix C.4), the results presented here serve as a conservative lower bound. The empirical difference between these adjacent configurations is marginal and does not alter the broader conclusions regarding the gate's ability to generalise.

Scaling to deep backbones. Both gates successfully protect past knowledge in harder tasks, though their visual impact varies by normalisation layer. Under batch normalisation (Figure 6.4(a)), the vanilla baseline suffers a moderate drop in *depth* and accumulates forgetting *area* as it slowly readapts. Although both gates still exhibit a minor performance dip, which we attribute to the shifting of running batch statistics rather than structural forgetting, they successfully intercept the primary degradation, effectively maintaining the past task near its peak performance.

The contrast is drastically sharper under group normalisation (Figure 6.4(b)). Here, the vanilla baseline suffers a catastrophic, prolonged crash, heavily inflating both the gap *depth* and *area*. Both the scalar and directional gates suppress this massive transient, keeping the trajectory virtually flat through the switch. As Table 6.2 demonstrates, this intervention vastly improves worst-case task retention (*min-ACC*) across the board without sacrificing final *accuracy*. The one exception is a three-task group-norm sequence, where the curriculum's gradient-norm ratio loses its conditioning at a boundary between two near-identical corruptions and the run destabilises to 32.8 ± 14.5 pp of depth against vanilla's 4.4 ± 0.4 (Appendix C.7).

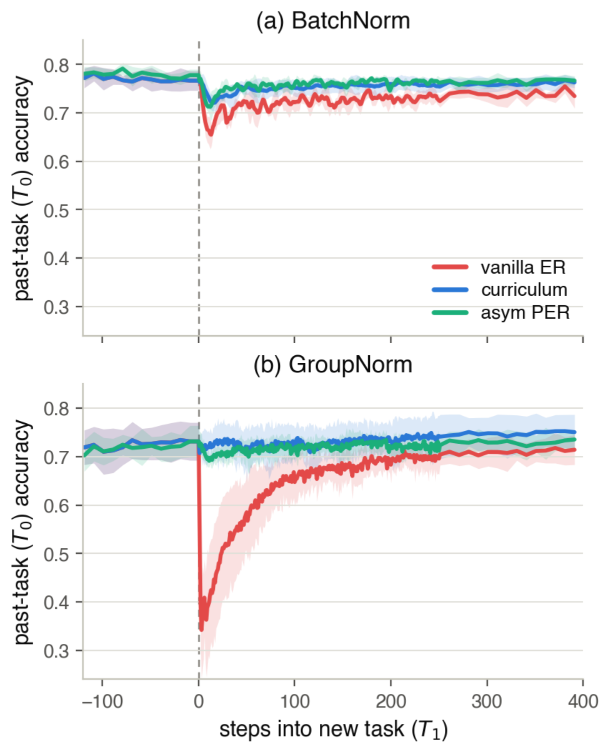


Figure 6.4: Per-step past-task (T_0) accuracy across the CIFAR-10 transition (ResNet-18, momentum 0.9, buffer 5,000), one panel per normalisation, each overlaying vanilla ER (red), the practical curriculum (blue) and asymmetric PER (aqua); seed mean with ±std bands, x=0 the task switch. (a) BatchNorm: vanilla ER dips and readapts slowly while both gates hold T_0 near its plateau. (b) GroupNorm: vanilla ER's transient is deep and long, whereas both gates hold T_0 flat through the switch (Table 6.2, Appendix C.6).

The deciding factor is compute cost. Since both methods successfully minimise the gap *depth* and *area* during the task transition, the deciding factor in deep networks becomes efficiency. Because the directional gate requires per-step conjugate-gradient solves to compute the filter, it consumes substantially more wall-clock time and GPU resources. The scalar gate, requiring only a single gradient-norm ratio, provides equivalent stability at a fraction of the computational footprint (Appendix C.8).

Norm	Method	gap_depth (pp)	gap_area_end	ACC	min-ACC
BatchNorm	G1: Vanilla ER	11.8 ± 2.5	17.5 ± 4.6	0.723 ± 0.008	0.644 ± 0.023
	G2: Curriculum ($\lambda_{\min}$=0.20)	$\mathbf{4.8 \pm 0.7}$	$\mathbf{3.6 \pm 2.2}$	0.722 ± 0.011	0.713 ± 0.017
	G3: Asym. PER (δ=0.1)	5.9 ± 4.3	3.3 ± 3.0	0.715 ± 0.011	0.698 ± 0.023
GroupNorm	G4: Vanilla ER	41.2 ± 13.5	34.9 ± 23.7	0.730 ± 0.011	0.309 ± 0.111
	G5: Curriculum ($\lambda_{\min}$=0.20)	$\mathbf{5.2 \pm 2.1}$	$\mathbf{1.7 \pm 0.9}$	0.731 ± 0.020	0.670 ± 0.055
	G6: Asym. PER (δ=0.1)	5.6 ± 1.1	4.8 ± 3.8	0.741 ± 0.009	0.673 ± 0.023

Table 6.2: CIFAR-10 headline block (ResNet-18, momentum 0.9, buffer 5,000, five seeds). Both gates cut gap depth and area far below vanilla ER at matched final accuracy and much higher worst-case retention (min-ACC), under both normalisations.

7 Discussion

The empirical results substantiate the two-level account of the stability gap and, by extension, the taxonomy of replay gating mechanisms. This section synthesizes these findings, contextualizes them within existing theory, acknowledges the limitations of the experimental design, and outlines implications for future method design.

7.1 Why the Gap Needs Two Levels

Our empirical results support a central conceptual claim: the stability gap is not a monolithic phenomenon, but rather the composition of two distinct optimization deviations that require separate interventions. The step-size ladder (Section 6.1) provides direct evidence for this distinction. Because uniformly rescaling the learning rate η acts as a time reparameterization of the gradient flow, it isolates how faithfully the trajectory is followed without altering the intended trajectory itself. When both Level-Two drivers are neutralized, by driving the step size toward zero and utilizing exact replay gradients, the performance drop does not vanish; instead, the underlying "arc" remains as a structural limit.

Conversely, at the standard operating point, the scalar curriculum repairs the trajectory (Level One) and mitigates the imbalance at the task boundary, yet a chronic, jittery residual remains. The transient gap only closes entirely when this path correction is combined with the removal of the remaining Level-Two driver: the estimator variance of the finite buffer (as demonstrated by C8, Section 6.2). This distinction also clarifies the role of the boundary imbalance. While the imbalance itself is a structural property of the Level-One loss landscape, the step size that interacts with it acts as the acute Level-Two driver that triggers the spike. As the curriculum introduces the new task slowly, it effectively shortens the step size taken across this imbalanced terrain, thereby suppressing the acute Level-Two trigger while simultaneously allowing the Level-One path intervention to stabilize the overall trajectory.

Crucially, this two-level framework disentangles two costs that are typically conflated. The permanent reduction in past-task performance represents the fundamental stability–plasticity trade-off. In contrast, the temporary transient drop, the part revealed solely by continual evaluation, represents a failure to follow the reference path. The stability gap is an artifact of this transient failure, not the inherent price of learning a new task.

Finally, this perspective explains why the dominant evaluation metric depends on the network architecture. On an MLP, the network reconverges within a few steps; the trajectory arc integrates to a negligible area, making the overshoot spike (*depth*) the primary indicator of severity. On a ResNet, the transient remains open for hundreds of iterations (Figure 6.4), meaning the slow recovery tail accumulates a massive loss in *area*. In this regime, both gates demonstrate their largest margin over vanilla ER (Table 6.2). Depth and area serve as the respective empirical signatures of Level Two and Level One, demonstrating that relying solely on depth metrics would severely understate the stability gap on deeper architectures where the continual-evaluation cost is highest.

7.2 Two Gates, Two Ways to Divide the Work

Within the unified update framework (Section 3.3), the curriculum and asymmetric PER represent two distinct paradigms: a scalar gate and a directional gate. The empirical results demonstrate that these gates succeed and fail exactly where the taxonomy predicts, governed primarily by *where* the attenuation is applied to the gradient update.

The scalar gate modulates the gradient at its source. By scaling the entire new-task loss via a schedule that initiates near zero, it restricts the magnitude of the push before discretization, momentum accumulation, or gradient noise can act upon it. Three consequences follow. First, it composes constructively with momentum, because the accumulator never registers a sustained, full-strength update to rebuild (Section 6.2). Second, it has no "blind spots" regarding the loss landscape curvature, because the adaptive schedule reacts to a live damage signal rather than relying on a precomputed model of the past task. Finally, it incurs a plasticity penalty: globally suppressing the loss inherently slows new-task learning, a delay that is only compensated for by an accelerator. The scalar gate trades plasticity for stability, and momentum serves as the mechanism to recover that plasticity.

Conversely, the directional gate operates on a per-step basis. By filtering the update through the replay Fisher matrix, it selectively decelerates learning only along directions flagged by past-task curvature. This protects past knowledge without a measured plasticity cost; both overall accuracy and short-horizon plasticity improve as the filter strengthens (Table C.5). However, this per-step locality introduces two distinct failure modes. The first is its vulnerability to momentum. Under momentum, the protective effect unravels via the re-inflation mechanism described in Section 6.3. The momentum accumulator integrates the localized, moderated updates over time, reconstructing the very displacement the filter was designed to prevent. This single distinction, whether attenuation occurs at the source or at the step, accounts for the entirely divergent behaviors of the two gates when momentum is applied.

The directional gate's second vulnerability is its susceptibility to delayed catastrophic forgetting, a failure paradoxically isolated to the exact-gradient controls. At the strongest filter setting, past-task accuracy can experience a sharp, late-stage cliff under a full buffer, a degradation entirely absent when using a standard finite buffer (Appendix C.4). At $\delta = 0.03$ it strikes 3 of 5 seeds after some two hundred flat steps, lifting mean depth to 9.5 ± 8.2 pp (Figure C.5). This indicates a latent deficiency in the curvature model: the sampled Fisher matrix may rate a direction as flat, allowing new-task gradient pressure to escape through it. While the sampling noise of a practical buffer inadvertently protects against this by preventing the iterate from settling into the escape direction, this exposure reveals a fundamental weakness of model control: it is only as robust as its static model of the past task. A feedback gate avoids this by reading the damage as it occurs.

Because the scalar gate matches the directional gate's retention without suffering from curvature blind spots, and avoids the $\sim16\times$ wall-clock and $\sim24\times$ energy overhead associated with per-step Fisher solves (Table C.10), it is the more practical solution under momentum. Consequently, $C7_M$ (the adaptive schedule with $\lambda_{\min} = 0.20$ under momentum) is the configuration this thesis recommends and the one successfully scaled to CIFAR-10.

7.3 Implications for the Literature

The two-level decomposition allows existing, seemingly disparate accounts of the stability gap to compose rather than compete. The boundary imbalance identified by De Lange et al. [5], the persistence under the exact joint loss demonstrated by Hess et al. [13], and the loss geometry derived by Kao et al. [16] describe different facets of a single overarching mechanism. The step-size ladder contextualizes these contributions. The imbalance is a structural feature of the intended trajectory, but it does not inherently produce the measured drop: the imbalance remains identical at every rung. It is the step size that converts this imbalance into a discrete drop, aligning with the optimizer-centric perspective of Obis [23]. Furthermore, the persistence of the drop under exact gradients represents the inherent "arc" of the trajectory. This decomposition connects the imbalance account to empirical measurements via the step size, and situates the geometric account as the fundamental limitation that no step-size adjustment can resolve.

This decomposition also repositions the evaluation of replay memory. Traditionally, buffer policies are judged by how well the stored set represents past data, typically scored at the end of a task where even small reservoirs appear competitive [1, 2, 4]. However, the continual evaluation lens exposes a second, more critical role for the buffer: the per-step fidelity of the restoring gradient it produces. A representative buffer does not guarantee a reliable per-step gradient, and this estimator variance acts as a chronic Level-Two driver of the stability gap. Crucially, because gradient unreliability reaches the iterate multiplied by the learning rate, buffer quality and step size operate on a single coupled axis. A memory budget is therefore only adequate relative to the step size it accompanies. Validating buffer policies at a single learning rate or solely via end-of-task accuracy obscures this dynamic.

Additionally, the unified update framework reclassifies existing replay methods based on where and when they apply modulation, cutting across conventional categorizations. For example, viewed through this formulation (Equation 2.1), methods like A-GEM [3] and MEGA [10] do not act as directional filters, nor do they attenuate the new-task gradient at its source. Instead, they dynamically scale the replay gradient in response to detected conflicts or loss ratios, adding counter-force rather than suppressing the initial push. The shared family is therefore scalar replay re-weighting, fundamentally separating them from source-attenuation methods like the curriculum.

This reclassification yields precise, falsifiable predictions for the literature. Because the new-task gradient remains unattenuated at its source in both A-GEM and MEGA, the momentum accumulator will inevitably build full new-task velocity regardless of the reactive replay force. Consequently, our framework predicts these methods will not compose constructively with momentum; their per-step corrections risk being overwhelmed by accumulated velocity, mirroring the failure mode of the directional gate.

7.4 Limitations

Several methodological and scope limitations constrain these findings. First, the experimental design utilizes five random seeds per configuration. This sample size structurally precludes traditional statistical significance testing (e.g., achieving $p < 0.05$ via non-parametric methods, Section 5.1.2). Consequently, the analysis relies on the magnitude of effect sizes and unanimous seed-paired directional consistency. However, the primary phenomena supporting our conclusions, such as the up-to-eightfold reduction in stability gap depth on CIFAR-10, the variance collapse between the C7 and C8 configurations, and the momentum-induced re-inflation of the directional gate, are vast in magnitude and uniformly present across all observed seeds.

Second, the magnitude of the trajectory arc was primarily quantified using a two-task rot-MNIST sequence on an MLP backbone. While this environment cleanly isolates the fundamental mechanics, the substantially longer transient recovery observed on the ResNet architecture (Figure 6.4) strongly suggests that the proportional balance between Level-One (trajectory) and Level-Two (variance/step-size) dynamics is highly architecture-dependent. Consequently, these two-level dynamics must be systematically verified in significantly larger models. As network capacity, width, and depth scale, the geometry of the loss landscape and the optimizer's traversal through it may behave fundamentally differently, potentially altering how the stability gap manifests and how the required interventions distribute across the two levels in state-of-the-art deep architectures.

The proposed gating mechanisms also exhibit specific limitations. The scalar gate's ability to preserve final accuracy is strictly contingent on the combined curriculum-and-momentum configuration; at zero momentum, extending the curriculum ramp successfully reduces the gap depth but incurs a permanent plasticity penalty (Table C.2). Additionally, the gradient-ratio feedback signal exhibits conditional fragility, losing proper conditioning when transitioning between two nearly identical data corruptions (Appendix C.7). For the directional gate, while solver controls verify that the late-stage performance cliffs are not numerical artifacts, the proposed curvature "blind spot" mechanism remains a hypothesized interpretation rather than a definitively proven cause.

Finally, the experimental scope is strictly limited by its treatment of the memory axis, which was sampled at only two extremes: a standard 1k reservoir and the full dataset for exact gradients. Furthermore, the evaluation relies on only a single memory management method (uniform reservoir sampling). Because buffer estimator variance acts as a primary Level-Two driver of the stability gap, evaluating only these two extremes leaves the intermediate dynamics unmapped. Consequently, the current results do not capture how a broader spectrum of memory budgets or intelligent core-set selection policies [1, 2] might shape the gradient distribution to modulate the variance term independently of the step size. Additionally, the task structure itself presents a limitation. Our primary analysis focuses narrowly on domain-incremental shifts within a single-boundary, two-task design. Although the five-task rot-MNIST sequence (Appendix C.9) indicates that these dynamics persist across multiple task boundaries, the absence of longer, class-incremental learning scenarios restricts the immediate generalizability of the two-level account across more complex continual learning environments.

7.5 Outlook

The proposed taxonomy highlights a missing paradigm: a directional gate driven by a live feedback signal rather than a precomputed curvature model. Such a mechanism would filter the update directionally, preserving the plasticity that makes feedforward gates efficient, while dynamically reading damage as it occurs, thereby avoiding the blind spots of static curvature models. Regardless of the implementation, the results under momentum strictly constrain future designs: any directional attenuation must act at the source or on the accumulated velocity, rather than on the instantaneous step, to avoid the re-inflation dynamics observed in Section 6.3.

Four specific avenues for future research emerge from these findings:

- *Adaptive and Non-Uniform Step Sizes.* While the step-size ladder maps the uniform step-size axis, the role of per-parameter adaptive optimizers [23] remains an open question. Crucially, if an optimizer applies directional attenuation to the instantaneous step, it must be determined whether it inherits the momentum-induced re-inflation observed here, given that these optimizers maintain internal accumulators.

- *Re-inflation in Existing Methods.* Any approach that preconditions the update using past-task curvature per step—including the natural-gradient family [16]—shares this vulnerability to momentum. Moving the attenuation from the per-step gradient to the accumulated velocity presents a directly testable modification for these architectures.

- *Efficient Directional Gates.* The computational burden of per-step conjugate-gradient solves could be mitigated by utilizing diagonal or Kronecker-factored Fisher approximations, refreshed sparsely at task boundaries. Determining whether the gate's protection survives these approximations is a prerequisite for scaling directional methods to architectures substantially larger than ResNet-18.

- *Alternative Variance Reducers.* Because momentum's role in closing the transient gap proved interchangeable with source variance reduction (C8), the scalar curriculum could be paired with alternative variance-reduction techniques. Iterate averaging, which avoids momentum's accumulator-driven overshoot entirely, is a natural candidate for integration.

Finally, the dynamics observed throughout these experiments demonstrate that the stability gap can no longer be treated as a singular artifact to be suppressed. It is fundamentally split between the structural trajectory dictated by the loss landscape (Level One) and the discrete overshoot introduced by the optimizer (Level Two). Future interventions must therefore move beyond blunt attenuation. By deliberately addressing both the underlying geometry of the task transition and the mechanics of the optimizer's traversal through it, the two-level framework presented here provides a blueprint for engineering more robust continual learning systems.

Statement on the Use of AI Tools

In accordance with the University of Groningen's guidelines on the responsible use of generative AI, I disclose that AI assistants were used during the preparation of this thesis. The tools were Google's Gemini 3.x series, Anthropic's Claude 4.x series, and Anthropic's Claude Fable 5, the releases current across the period, February to July 2026, in which the work was carried out. They functioned as assistive tools under my continuous direction, not as autonomous contributors. Specifically, I used them to brainstorm and stress-test ideas and as a sounding board for reasoning through arguments and explanations; to draft and revise prose, which I subsequently edited for accuracy, precision, and voice; to assist in writing and refactoring the analysis and experiment code, all of which I reviewed, ran, and tested; and to produce first-pass summaries of the literature, which I verified against the primary sources.

The tools were never given open-ended control: every request specified a concrete, bounded task, and all output was reviewed and, where retained, corrected, understood, and integrated by me. All experimental results, figures, and numerical claims were checked against the underlying run data, and all cited work was read in the original. The research questions, experimental design, interpretation of the results, and the overall argument of the thesis are my own, and I take full responsibility for its content, including any errors.

References

[1] Rahaf Aljundi, Min Lin, Baptiste Goujaud, and Yoshua Bengio. Gradient based sample selection for online continual learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. URL https://arxiv.org/abs/1903.08671. 3, 4, 16, 17

[2] Jihwan Bang, Heesu Kim, YoungJoon Yoo, Jung-Woo Ha, and Jonghyun Choi. Rainbow memory: Continual learning with a memory of diverse samples, 2021. URL https://arxiv.org/abs/2103.17230. 16, 17

[3] Arslan Chaudhry, Marc'Aurelio Ranzato, Marcus Rohrbach, and Mohamed Elhoseiny. Efficient lifelong learning with a-gem. In *International Conference on Learning Representations (ICLR)*, 2019. URL https://arxiv.org/abs/1812.00420. 4, 16

[4] Arslan Chaudhry, Marcus Rohrbach, Mohamed Elhoseiny, Thalaiyasingam Ajanthan, Puneet K. Dokania, Philip H. S. Torr, and Marc'Aurelio Ranzato. On tiny episodic memories in continual learning, 2019. URL https://arxiv.org/abs/1902.10486. 4, 16

[5] Matthias De Lange, Gido van de Ven, and Tinne Tuytelaars. Continual evaluation for lifelong learning: Identifying the stability gap. In *International Conference on Learning Representations (ICLR)*, 2023. URL https://arxiv.org/abs/2205.13452. 3, 4, 6, 8, 9, 16, 24

[6] Felix Draxler, Kambis Veschgini, Manfred Salmhofer, and Fred A. Hamprecht. Essentially no barriers in neural network energy landscape, 2019. URL https://arxiv.org/abs/1803.00885. 5

[7] Jonathan Frankle, Gintare Karolina Dziugaite, Daniel M. Roy, and Michael Carbin. Linear mode connectivity and the lottery ticket hypothesis, 2020. URL https://arxiv.org/abs/1912.05671. 5

[8] Timur Garipov, Pavel Izmailov, Dmitrii Podoprikhin, Dmitry Vetrov, and Andrew Gordon Wilson. Loss surfaces, mode connectivity, and fast ensembling of dnns. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018. URL https://arxiv.org/abs/1802.10026. 5

[9] Yiduo Guo, Jie Fu, Huishuai Zhang, Dongyan Zhao, and Yikang Shen. Efficient continual pretraining by mitigating the stability gap, 2024. URL https://arxiv.org/abs/2406.14833. 4

[10] Yunhui Guo, Mingrui Liu, Tianbao Yang, and Tajana Rosing. Improved schemes for episodic memory-based lifelong learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. URL https://arxiv.org/abs/1909.11763. 4, 16

[11] Md Yousuf Harun and Christopher Kanan. Overcoming the stability gap in continual learning. In *Conference on Lifelong Learning Agents (CoLLAs)*, 2024. URL https://arxiv.org/abs/2306.01904. 4

[12] Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In *International Conference on Learning Representations (ICLR)*, 2019. URL https://arxiv.org/abs/1903.12261. 8, 21

[13] Timm Hess, Tinne Tuytelaars, and Gido M. van de Ven. Two complementary perspectives to continual learning: Ask not only what to optimize, but also how. In *Conference on Lifelong Learning Agents (CoLLAs)*, 2024. URL https://arxiv.org/abs/2311.04898. 3, 4, 16

[14] Shun ichi Amari. Natural gradient works efficiently in learning. *Neural Computation*, 10(2): 251–276, 1998. 7

[15] Sandesh Kamath, Albin Soutif-Cormerais, Joost van de Weijer, and Bogdan Raducanu. The expanding scope of the stability gap: Unveiling its presence in joint incremental learning of homogeneous tasks, 2024. URL https://arxiv.org/abs/2406.05114. 4

[16] Ta-Chu Kao, Kristopher T. Jensen, Gido M. van de Ven, Alberto Bernacchia, and Guillaume Hennequin. Natural continual learning: success is a journey, not (just) a destination. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. URL https://arxiv.org/abs/2106.08085. 3, 4, 6, 7, 16, 17

[17] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114 (13):3521–3526, March 2017. ISSN 1091-6490. doi: 10.1073/pnas.1611835114. URL http://dx.doi.org/10.1073/pnas.1611835114. 4

[18] David Lopez-Paz and Marc'Aurelio Ranzato. Gradient episodic memory for continual learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. URL https://arxiv.org/abs/1706.08840. 4

[19] James Martens. New insights and perspectives on the natural gradient method. *Journal of Machine Learning Research*, 21(146):1–76, 2020. URL http://jmlr.org/papers/v21/17-678.html. 23

[20] Michael Mccloskey and Neil J. Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. *The Psychology of Learning and Motivation*, 24:104–169, 1989. 3

[21] Martial Mermillod, Aurélia Bugaiska, and Patrick Bonin. The stability-plasticity dilemma: Investigating the continuum from catastrophic forgetting to age-limited learning effects. *Frontiers in psychology*, 4:504, 08 2013. doi: 10.3389/fpsyg.2013.00504. 3

[22] Seyed Iman Mirzadeh, Mehrdad Farajtabar, Dilan Gorur, Razvan Pascanu, and Hassan Ghasemzadeh. Linear mode connectivity in multitask and continual learning. In *International Conference on Learning Representations (ICLR)*, 2021. URL https://arxiv.org/abs/2010.04495. 5

[23] Chris Obis. Reaching for resilience: Understanding how optimizers affect the stability gap in continual learning. Bachelor thesis, Delft University of Technology, 2025. URL https://repository.tudelft.nl/file/File_356b427b-c259-4ad5-b47c-16cc7c234716. Supervised by T. Viering and G. M. van de Ven. 4, 6, 16, 17

[24] Roger Ratcliff. Connectionist models of recognition memory: Constraints imposed by learning and forgetting functions. *Psychological Review*, 97:285–308, 04 1990. doi: 10.1037/0033-295X.97.2.285. 3

[25] Nicol N. Schraudolph. Fast curvature matrix-vector products for second-order gradient descent. *Neural Computation*, 14(7):1723–1738, 2002. doi: 10.1162/08997660260028683. 23

[26] Shagun Sodhani, Mojtaba Faramarzi, Sanket Vaibhav Mehta, Pranshu Malviya, Mohamed Abdelsalam, Janarthanan Janarthanan, and Sarath Chandar. An introduction to lifelong supervised learning, 2022. URL https://arxiv.org/abs/2207.04354. 3, 4

[27] Albin Soutif-Cormerais, Antonio Carta, and Joost Van de Weijer. Improving online continual learning performance and stability with temporal ensembles, 2023. URL https://arxiv.org/abs/2306.16817. 4

[28] Edoardo Urettini and Antonio Carta. Online curvature-aware replay: Leveraging 2nd order information for online continual learning, 2025. URL https://arxiv.org/abs/2502.01866. 7

[29] Gido M. van de Ven, Nicholas Soures, and Dhireesha Kudithipudi. *Continual learning and catastrophic forgetting*, page 153–168. Elsevier, 2025. ISBN 9780443157554. doi: 10.1016/b978-0-443-15754-7.00073-0. URL http://dx.doi.org/10.1016/B978-0-443-15754-7.00073-0. 4

[30] Jeffrey S. Vitter. Random sampling with a reservoir. *ACM Trans. Math. Softw.*, 11(1):37–57, March 1985. ISSN 0098-3500. doi: 10.1145/3147.3165. URL https://doi.org/10.1145/3147.3165. 4, 22

A Implementation Details

This appendix records the implementation specifics that the methods of Section 5 depend on but that would interrupt the argument in the main text: the datasets, the backbones and training protocol, the experience-replay variants, and the asymmetric-PER solver. The full implementation is available at https://github.com/Arthurreuss/ValleyWalk.

A.1 Datasets

rot-MNIST rotates each 28×28 image by the per-task angle and feeds the flattened greyscale vector to the MLP. The CIFAR-10 domain shifts are generated with the `imagecorruptions` package, which implements the standard common-corruptions benchmark of Hendrycks and Dietterich [12]; the Gaussian noise, shot noise, and contrast shifts are applied at severity 3 on the benchmark's 1–5 scale, per sample at load time, followed by the standard CIFAR-10 channel normalisation. The clean task ("none") applies the normalisation only.

A.2 Models and Training

Tables A.1 and A.2 give the two backbones, and Table A.3 the training protocol shared across both benchmarks.

Component	Value
Input	784 (flattened 28×28 greyscale)
Hidden layers	2×400 (fully connected, ReLU)
Output	10-class logits
Regularisation	none
Total parameters	478,410

Table A.1: MLP backbone used for the rot-MNIST sweeps.

Component	Value
Input	$3 \times 32 \times 32$ RGB
Stem	3×3 conv, stride 1, no max-pool (CIFAR adaptation)
Stages	4 residual stages, BasicBlocks $[2, 2, 2, 2]$, widths $[64, 128, 256, 512]$
Normalisation	BatchNorm (default); GroupNorm ($\min(32, C)$ groups) variant crossed in Section 5.2.5
Head	global average pool + linear, 10-class logits
Total parameters	≈11.2M

Table A.2: CIFAR-adapted ResNet-18 backbone used for the CIFAR-10 block. The 3×3 stride-1 stem and removed max-pool keep the feature map at 32×32; all other details follow the ResNet-18.

Hyperparameter	rot-MNIST (MLP)	CIFAR-10 (ResNet-18)
Epochs per task	1 (online); c in the ladder	10
Batch size	256	256
Optimiser	SGD	SGD
Learning rate	0.1; $0.1/c$, $c \in \{1, 3, 10, 100\}$, in the ladder	0.1
Momentum	0.0 or 0.9	0.9 (0.0 in cross)
Weight decay	0.0	0.0
Replay buffer	1,000 or 60,000	5,000
Seeds	5	5 or 3

Table A.3: Training protocol for both benchmarks.

The per-step plotting convention. Test accuracy is logged every 10 steps away from a transition and every step inside the dense window (Section 5.1.1), so a short interval separates the last pre-switch evaluation from the first post-update one. In the per-step figures the last pre-switch level is held flat up to the boundary rather than interpolated across that interval: the past task is still training and sitting at its plateau there, so a straight line between the two samples would draw the drop as beginning before the switch that causes it. The convention affects the drawn curve only. Every reported metric is computed from the logged samples themselves, so none of them depends on it.

Drawing runs of different length on one axis. The rungs of the ladder (Section 5.2.1) are not the same length in steps: at matched flow time the epoch count scales as $1/\eta$, so the $\eta = 0.001$ rung takes a hundred times as many updates as the standard one to cover the same $\tau = 23.5$. Its figures are therefore drawn against flow time $\tau = \eta \cdot$ (steps into T_1) rather than against the step index, which is the axis on which a uniform rescaling of η is a reparametrisation of one and the same gradient flow. The evaluation cadence is scaled by the same factor $c = 0.1/\eta$ (Appendix A.4), so each rung contributes 235 samples at identical τ values and the curves are compared sample-for-sample rather than interpolated onto a common grid; areas convert with the same factor, $A_\tau = \eta A_{\text{steps}}$ (Section 5.1.2). Every block outside the ladder runs at a single η, where the two axes differ by a constant, and keeps the step index.

A.3 Experience replay

The replay buffer uses reservoir sampling [30], so the stored set is a uniform sample of the past stream at any budget. Two modes appear in the experiments:

- *Standard ER* (the D1 reference, the ladder's 1k arm, and all C-series conditions except C8): each step takes one combined backward pass on $L_{\text{current}} + L_{\text{replay}}$, with half the mini-batch drawn from the current task and half sampled with replacement from the buffer.
- *Full-data replay* (the ladder's exact arm, C8, and the P2–P3 legs): the replay gradient is computed on *every* stored sample exactly once per step (no sub-sampling). Paired with a buffer equal to the full past-task training set (60,000 for rot-MNIST T_0), this yields the deterministic empirical past-task gradient at the current parameters, removing estimator variance.

The λ-curriculum re-weights only the current-task term, minimising $L_\lambda = L_{\text{replay}} + \lambda(t)\, L_{\text{current}}$; the adaptive schedule reads $\|g_{\text{replay}}\|/\|g_{\text{new}}\|$ through an exponential moving average ($\alpha = 0.05$) and applies the optional floor $\lambda_{\min}$. The EMA is reset to zero at each task boundary, so λ starts near zero on every task; the time-indexed schedules latch at $\lambda = 1$ once the ramp is past, but the adaptive one has no such state and follows the ratio for the whole task.

Reconstructing λ from the logged gradient norms of the C4 runs shows how far the two momentum legs actually get, and they get nowhere near the same place. At momentum 0 the norms do converge over the epoch, the median $\|g_{\text{new}}\|/\|g_{\text{replay}}\|$ falling from $\approx$7 over the first twenty steps to $\approx$1.1 over the second half, and λ follows it to 0.79–0.93 at the end of the task; it reaches 1 in one seed of five and then only at step 208 of 235, so even here the release is incomplete when the task ends. At momentum 0.9 the ratio instead widens, from $\approx$10 early to $\approx$40 late, and λ peaks at 0.12–0.16 around step 60 and decays to $\approx$0.02: the gate shuts again rather than opening. With the floor of C5–C7 in place this shows up as saturation, C7$_M$ sitting at $\lambda = \lambda_{\min}$ for 73–79% of the task's steps. The schedule is a feedback rule with no terminal state, not a ramp with an endpoint, and the momentum-on results of Section 6.2 are obtained in the regime where it holds the new-task term down throughout.

A.4 The step-size ladder

Every cell of the ladder warm-starts from a shared task-0 checkpoint. `init.checkpoint` loads a task-0 `state_dict` written by a separate anchor run, one per momentum setting and seed, trained at the standard operating point ($\eta = 0.1$, one epoch); `init.skip_tasks` then suppresses the training loop of that many leading tasks while still running their `end_task` bookkeeping, so the replay buffer is filled from the skipped task's stream exactly as it would have been in an ordinary run. Because reservoir sampling at a fixed seed returns the same subset of the same stream at a given budget, θ_0^* and the buffer are identical across the four rungs at a given buffer arm, momentum, and seed, and the learning rate is the only difference between them.

A skipped task still advances `global_step` by one nominal epoch ($\lceil 60{,}000/256 \rceil = 235$ batches), so the task boundary carries the same index as in an ordinary run and the per-step plotting convention above keys off it unchanged. One pre-switch evaluation is written at step 234, supplying the flat pre-boundary segment that the suppressed training loop would otherwise have recorded.

Inside the dense post-switch window the stability-gap cadence fires on the step index *within the current task* rather than on the global step. At a cadence of $c > 1$ a global-step modulus would place the first post-switch sample at an arbitrary phase offset into the task whenever c does not divide the 235-batch epoch, skipping exactly the steps the boundary spike occupies. The change is a no-op at cadence 1, which is every previously archived run.

The four rungs use $\eta = 0.1$, $\eta = 0.1/3 = 0.0\overline{3}$, $\eta = 0.01$ and $\eta = 0.001$. The second is written into the configuration at full precision rather than rounded, so that $\eta c = 0.1$ holds exactly and the flow time $\tau = 23.5$ is matched to the last digit; the main text reports it as 0.1/3. The dense window is set to $250c$ steps, which exceeds the $235c$ steps of the task, so every cell records the whole of T_1 at the dense cadence: 235 samples per rung, evenly spaced in flow time. Buffer-fidelity diagnostics are disabled throughout the ladder, since each such step costs a full pass over the past-task training set and the longest rung takes one hundred epochs.

A.5 Asymmetric PER

How the Fisher is computed. For a softmax cross-entropy model the Fisher information matrix equals the generalised Gauss–Newton matrix

$$F \;=\; \frac{1}{B}\sum_{b} J_b^\top H_b J_b, \qquad J_b = \frac{\partial z_b}{\partial \theta}, \qquad H_b = \mathrm{diag}(p_b) - p_b p_b^\top,$$

where z_b are the logits of sample b and $p_b = \mathrm{softmax}(z_b)$ [19]. This matrix is positive semi-definite by construction, so $F + \delta I$ is positive-definite and the solve is well-posed; it depends only on the model's predictive distribution, not on the labels. F_{rep} is estimated on a mini-batch sampled from the replay buffer and *recomputed at every step from the current parameters*: the gate uses a live local curvature model, not a snapshot frozen at the task boundary. It remains feedforward in the sense of Section 3.3, because what it reads is the curvature model, never the past-task performance it is meant to protect. The matrix is never formed explicitly. Fisher-vector products $v \mapsto Fv$ are computed matrix-free by the Gauss–Newton trick of Schraudolph [25]: one forward pass gives the logits (retained across the solve), each product then costs one Jacobian-vector and one vector-Jacobian product.

The solve. The filtered push $\delta\,(F_{\mathrm{rep}} + \delta I)^{-1} g_{\mathrm{cur}}$ of Equation (3.4) is computed each step by conjugate gradients over these Fisher-vector products (25 iterations, relative-residual tolerance 10^{-4}), warm-started from the previous step's solution, which keeps the per-step solve cheap across the run; the warm start is dropped at task boundaries, where the landscape shifts. The solver control of Section 5.2.4 reruns the strongest filter ($\delta = 0.03$, full buffer) at 60 iterations to separate properties of the gate from artefacts of an under-converged solve. On the full-buffer leg the replay gradient is exact over all 60k past samples while the Fisher stays estimated on a sampled batch, since a full-buffer Fisher-vector product per CG iteration would dominate the runtime.

Order of gate and momentum. When momentum is enabled, the gate is applied first: the filtered direction d is written in place of the gradient, and the optimiser's velocity buffer accumulates $v \leftarrow \mu v + d$ with the step $-\eta\, v$. The velocity is therefore built from already-filtered directions and is itself applied unfiltered. The alternative ordering, accumulating raw gradients and filtering the velocity, is a distinct design with potentially different behaviour under the re-inflation mechanism of Section 6.3; it is not implemented here and is discussed as future work in Section 7.5.

B Metric Definitions

This appendix gives the full definitions of every metric recorded by the evaluation pipeline of Section 5.1.2. The main text explains the measures the argument turns on; the tables below are the look-up reference for the complete set.

Stability-gap metrics. Computed from the dense per-step evaluation. Write a_j^{pre} for the pre-task accuracy on task j and $a_j(t)$ for its accuracy at step t of the new task.

Metric	Definition	Role
`gap_depth`	$\max_j \left(a_j^{\text{pre}} - \min_t a_j(t)\right)$ over the full record window	Primary metric, worst single point of instability
`gap_area_end`	$\sum_j \int_0^T \max(0, b_j - a_j(t))\,dt$, trapezoidal over the full new-task record (to task end T), with $b_j = \min\left(a_j^{\text{pre}}, \bar{a}_j^{\text{end}}\right)$ and $\bar{a}_j^{\text{end}}$ the trailing-window mean (last $\max(5, \lceil 0.1n \rceil)$ samples); accuracy at or above b_j contributes 0	Transient cost reported in the results: dip below the recovered level until it closes, excluding permanent forgetting and continued learning (units: accuracy · steps)
`recovery_steps`	The first step (counted from the transition) at which *every* past task is simultaneously back at $\geq 90\%$ of its pre-task baseline; undefined (reported as "—") if no sub-threshold dip occurs, or if recovery is not reached within the record	Duration of the transient, how long the gap stays open

Table B.1: Stability-gap metrics, recorded from dense per-step evaluation. `gap_depth` and `gap_area_end` are the two axes used in the main analysis, how deep the gap cuts and how long it stays open, together with `recovery_steps`.

Continual-learning metrics. Following De Lange et al. [5], computed at task boundaries and over the full record. Let $A(E_j, f_n)$ be the accuracy on the test set of task j evaluated at the model f_n after training stage n, and let N be the number of tasks. The full task-boundary accuracy matrix $R[i,j] = A(E_j, f_{T_i})$ is logged for every run, so any downstream metric can be reconstructed.

Metric	Definition	Role
ACC	$\frac{1}{N}\sum_j A(E_j, f_{T_{N-1}})$	Average task accuracy at the final model
FORG	$\frac{1}{N-1}\sum_{i<N-1}\left[A(E_i, f_{T_i}) - A(E_i, f_{T_{N-1}})\right]$	Average drop from right-after-learning to final
min-ACC	$\frac{1}{N-1}\sum_{i<N-1}\min\{A(E_i, f_n) : n > T_i\}$	Worst-case retention since learning
WF_w	Mean over tasks of the largest accuracy drop in any window of w consecutive evaluations ($w \in \{10, 100\}$)	Worst-case stability over short / medium horizons
WP_w	Symmetric counterpart of WF_w: the largest accuracy gain ($w \in \{10, 100\}$)	Plasticity / learning velocity
WC-ACC	$\frac{1}{N}A(E_{N-1}, f_{T_{N-1}}) + \left(1 - \frac{1}{N}\right)\text{min-ACC}$	Worst-case trade-off, a lower bound on ACC

Table B.2: Continual-learning metrics of De Lange et al. [5], recorded at task boundaries and over the full record.

Gradient-fidelity diagnostics. Recorded at the dense cadence on the vanilla-ER reference runs only. The tracker compares the replay-buffer gradient g_{replay} against the true past-task gradient g_{true}, computed at every step by a separate forward and backward pass over the entire past-task training set (no sub-sampling, so g_{true} is the deterministic empirical past-task gradient at the current parameters).

Diagnostic	Logged name	Role
$\cos(g_{\text{replay}}, g_{\text{true}})$	`true_grad_cosine`	Buffer-side directional fidelity, the estimator-variance driver of Section 5.2.1
$\lVert g_{\text{replay}}\rVert / \lVert g_{\text{true}}\rVert$	`true_grad_mag_ratio`	Buffer-side magnitude fidelity
$\lVert g_{\text{new}}\rVert / \lVert g_{\text{replay}}\rVert$	`grad_ratio`	Update-side magnitude asymmetry, large at step 1 when $\lVert g_{\text{replay}}\rVert \approx 0$

Table B.3: Gradient-fidelity diagnostics. Each is logged as a per-step series together with its mean (and minimum, for the cosine). The first two carry the `true_grad` prefix because they are measured against g_{true}; `grad_ratio` compares g_{replay} to the current-task gradient g_{new} and its inverse drives the adaptive curriculum of Section 5.2.2.

C Complete Results

This appendix gives the full metric set for every condition of the main experimental blocks, as recorded by the aggregation pipeline of Section 5.1.2. Each row is the mean over five seeds (three for the CIFAR generalisation block) with the seed standard deviation. Accuracy-type metrics (ACC, FORG, min-ACC, WF_{100}, WP_{100}, WC-ACC) are fractions; `gap_depth` is in percentage points (pp); `gap_area_end` is the transient cost (accuracy-fraction $\times$ steps, Table B.1), the dip below the level each task recovers to, so permanent forgetting does not contribute; "Recov." is `recovery_steps`, with "—" marking conditions where every past task stayed above the 90% recovery threshold for the whole window (no recovery event to time).

C.1 rot-MNIST: The Step-Size Ladder (S-series)

The sixteen cells of the ladder of Section 5.2.1: four rungs at matched flow time $\tau = 23.5$, each on the 1k reservoir and on the 60k exact past-task gradient, at both momentum settings, all warm-started from one frozen θ_0^* per momentum setting and seed.

Condition	ACC	FORG	min-ACC	WF_{100}	WP_{100}	WC-ACC	Depth	A_τ
Momentum 0.0								
S1 η=0.1 (1k)	0.874 ± 0.008	0.020 ± 0.019	0.653 ± 0.035	0.153 ± 0.017	0.454 ± 0.020	0.770 ± 0.019	22.9 ± 4.2	0.577 ± 0.080
S1 η=0.1 (exact)	0.895 ± 0.002	-0.020 ± 0.014	0.658 ± 0.044	0.138 ± 0.026	0.453 ± 0.025	0.773 ± 0.022	22.3 ± 5.0	0.590 ± 0.172
S2 η=0.1/3 (1k)	0.876 ± 0.004	0.021 ± 0.012	0.826 ± 0.007	0.036 ± 0.004	0.365 ± 0.005	0.858 ± 0.004	5.6 ± 1.6	0.293 ± 0.044
S2 η=0.1/3 (exact)	0.897 ± 0.002	-0.021 ± 0.012	0.839 ± 0.002	0.029 ± 0.002	0.369 ± 0.006	0.865 ± 0.002	4.2 ± 1.2	0.360 ± 0.167
S3 η=0.01 (1k)	0.877 ± 0.002	0.020 ± 0.013	0.832 ± 0.006	0.031 ± 0.004	0.368 ± 0.005	0.862 ± 0.003	4.9 ± 1.4	0.275 ± 0.055
S3 η=0.01 (exact)	0.897 ± 0.002	-0.020 ± 0.012	0.842 ± 0.003	0.025 ± 0.002	0.374 ± 0.003	0.867 ± 0.002	3.9 ± 1.2	0.346 ± 0.164
S4 η=0.001 (1k)	0.877 ± 0.002	0.020 ± 0.013	0.834 ± 0.006	0.028 ± 0.004	0.369 ± 0.006	0.863 ± 0.003	4.7 ± 1.4	0.270 ± 0.052
S4 η=0.001 (exact)	0.898 ± 0.001	-0.022 ± 0.012	0.843 ± 0.003	0.023 ± 0.002	0.376 ± 0.003	0.868 ± 0.002	3.8 ± 1.2	0.341 ± 0.163
Momentum 0.9								
$\mathrm{S1}_M$ η=0.1 (1k)	0.929 ± 0.003	0.058 ± 0.006	0.802 ± 0.023	0.087 ± 0.008	0.442 ± 0.006	0.881 ± 0.011	15.4 ± 2.2	0.105 ± 0.022
$\mathrm{S1}_M$ η=0.1 (exact)	0.965 ± 0.001	-0.011 ± 0.004	0.814 ± 0.011	0.076 ± 0.006	0.457 ± 0.009	0.889 ± 0.006	14.1 ± 1.2	0.301 ± 0.034
$\mathrm{S2}_M$ η=0.1/3 (1k)	0.936 ± 0.003	0.053 ± 0.003	0.855 ± 0.009	0.054 ± 0.004	0.436 ± 0.008	0.913 ± 0.004	10.1 ± 0.7	0.021 ± 0.005
$\mathrm{S2}_M$ η=0.1/3 (exact)	0.970 ± 0.001	-0.015 ± 0.003	0.857 ± 0.005	0.053 ± 0.002	0.454 ± 0.007	0.913 ± 0.003	9.9 ± 0.5	0.122 ± 0.014
$\mathrm{S3}_M$ η=0.01 (1k)	0.940 ± 0.002	0.048 ± 0.003	0.901 ± 0.005	0.029 ± 0.002	0.419 ± 0.006	0.937 ± 0.003	5.4 ± 0.3	0.003 ± 0.001
$\mathrm{S3}_M$ η=0.01 (exact)	0.972 ± 0.001	-0.017 ± 0.003	0.905 ± 0.005	0.027 ± 0.001	0.435 ± 0.007	0.939 ± 0.003	5.0 ± 0.3	0.061 ± 0.012
$\mathrm{S4}_M$ η=0.001 (1k)	0.942 ± 0.002	0.045 ± 0.002	0.910 ± 0.003	0.020 ± 0.001	0.411 ± 0.007	0.942 ± 0.002	4.6 ± 0.2	0.0006 ± 0.0001
$\mathrm{S4}_M$ η=0.001 (exact)	0.973 ± 0.001	-0.018 ± 0.003	0.928 ± 0.003	0.014 ± 0.002	0.426 ± 0.007	0.950 ± 0.002	2.7 ± 0.4	0.040 ± 0.011

Table C.1: Step-size ladder on two-task rot-MNIST, full metrics: rungs $\eta = 0.1$, $0.1/3$, 0.01 and 0.001 at 1, 3, 10 and 100 epochs, each on both buffer arms and at both momentum settings, five seeds. Depth is a maximum over accuracies and so is η-invariant. Area is *not*: `gap_area_end` is a trapezoidal integral of the drop curve against the step index, weighted by the spacing between records (units accuracy·steps, Section 5.1.2), so an identically shaped excursion reads $c = 0.1/\eta$ times larger on a rung stretched by c. The area column is therefore reported in flow time, $A_\tau = \eta\, A_{\text{steps}}$, which is the integral of the same curve against $\tau = \sum_t \eta$ and is comparable across rungs; the fixed-η blocks elsewhere in this appendix keep the step-axis value. WF_{100} and WP_{100} are fixed hundred-*step* horizons and therefore span a hundredth of the flow time at $\eta = 0.001$ that they span at $\eta = 0.1$; their movement down the ladder is a units artefact of the reparametrisation (Section 6.1).

Momentum is not simply a longer step. The grid provides two matched-effective-step diagonals, and together they say what one alone could not. At effective step 0.1, $(\eta = 0.01, \mu = 0.9)$ against $(\eta = 0.1, \mu = 0)$ separates widely, 5.0 pp / 0.061 / 0.972 against 22.3 pp / 0.590 / 0.895 on the exact arm; at effective step 0.01, $(\eta = 0.001, \mu = 0.9)$ against $(\eta = 0.01, \mu = 0)$ reads 2.7 pp / 0.040 / 0.973 against 3.9 pp / 0.346 / 0.897. The depth separation collapses from 17.3 to 1.2 pp as the matched step shrinks, which is what two legs approaching their own $\eta \to 0$ limits must do; the area separation does not, staying eightfold at the finest matched pair. What differs between the legs is therefore the shape of the residual excursion and not only its scale. This remains the secondary read Section 5.2.1 calls it: the legs warm-start from anchors trained at their own momentum and settle some seven accuracy points apart, so comparing their levels compares two operating points.

Momentum's arc is depth without tail. The within-leg contrast is both the safer comparison and the sharper one. On the sampled arm the momentum leg's flow-time area falls by more than two orders of magnitude across the ladder, $0.105 \to 0.0006$, and extrapolates to zero (-0.001), while its depth converges on 4.1 pp. In the limit the transient that survives under momentum is a dip that recovers almost at once: it keeps the instantaneous drop and loses the hanging tail. The $\mu = 0$ arc keeps both, 4.7 pp of depth and 0.27 of area that no further shortening of the step removes. Of the two-part measure of Section 5.1.2, then, momentum leaves essentially only the depth term standing.

The coarse rung reproduces the operating point in distribution. $\eta = 0.1$ on the sampled arm is the standard operating point, but it is not the archived vanilla-ER reference run (D1, Appendix C.2) and is not paired with it: the ladder skips T_0 training so that every cell can share one θ_0^*, which leaves the two protocols on different RNG streams and hence with different buffer contents seed for seed. Their distributions agree, 22.9 ± 4.2 against 19.5 ± 4.1 pp of depth and 5.77 ± 0.80 against 6.27 ± 0.47 of step-axis area, one-standard-deviation intervals overlapping on both. That is a check that warm-starting has not moved the operating point, and nothing stronger than that.

C.2 rot-MNIST: λ-Curriculum (C-series)

Vanilla ER at the standard protocol (D1: 1k reservoir, $\eta = 0.1$, one online epoch per task, the two gradients summed as they come) is the reference every rot-MNIST comparison is quoted against, and its full metric row heads each momentum block below rather than standing in a table of its own. These are the runs from which the buffer-fidelity diagnostics of Table C.4 (Appendix C.3) are taken; the step-size ladder re-runs its own $\eta = 0.1$ rung rather than reusing the row, because every cell of the ladder has to share one frozen θ_0^* (Appendix A.4).

Condition	ACC	FORG	min-ACC	WF_{100}	WP_{100}	WC-ACC	Depth	Area end	Recov.
Momentum 0.0									
Vanilla ER (baseline)	0.873 ± 0.005	0.019 ± 0.020	0.706 ± 0.056	0.147 ± 0.029	0.736 ± 0.020	0.794 ± 0.028	19.5 ± 4.1	6.3 ± 0.5	3.6 ± 0.9
C1 linear N50	0.851 ± 0.023	0.042 ± 0.021	0.707 ± 0.019	0.148 ± 0.011	0.748 ± 0.006	0.784 ± 0.021	17.5 ± 1.7	4.9 ± 0.5	35.8 ± 6.3
C2 linear N100	0.844 ± 0.025	0.046 ± 0.023	0.719 ± 0.033	0.140 ± 0.025	0.730 ± 0.006	0.786 ± 0.019	16.2 ± 4.3	3.6 ± 0.3	64.0 ± 8.6
C3 linear N200	0.828 ± 0.025	0.054 ± 0.024	0.739 ± 0.048	0.128 ± 0.043	0.688 ± 0.004	0.784 ± 0.034	14.3 ± 4.6	1.8 ± 0.6	132.8 ± 43.4
C4 adaptive	0.833 ± 0.025	0.050 ± 0.019	0.754 ± 0.025	0.111 ± 0.016	0.703 ± 0.007	0.794 ± 0.020	12.8 ± 2.9	2.2 ± 0.5	88.8 ± 6.1
C5 adaptive lmin0.05	0.835 ± 0.025	0.048 ± 0.019	0.749 ± 0.027	0.113 ± 0.016	0.702 ± 0.006	0.793 ± 0.020	13.2 ± 3.2	2.2 ± 0.5	88.8 ± 6.1
C6 adaptive lmin0.10	0.836 ± 0.026	0.048 ± 0.021	0.743 ± 0.026	0.120 ± 0.019	0.700 ± 0.007	0.791 ± 0.019	13.9 ± 3.3	2.4 ± 0.5	88.4 ± 11.0
C7 adaptive lmin0.20	0.837 ± 0.031	0.049 ± 0.024	0.743 ± 0.030	0.119 ± 0.023	0.710 ± 0.010	0.792 ± 0.024	13.9 ± 3.4	2.7 ± 0.6	64.8 ± 8.2
Momentum 0.9									
Vanilla ER (baseline)	0.928 ± 0.003	0.058 ± 0.007	0.797 ± 0.028	0.092 ± 0.016	0.808 ± 0.012	0.878 ± 0.014	15.9 ± 2.6	1.0 ± 0.2	12.6 ± 1.7
C1 linear N50	0.932 ± 0.003	0.053 ± 0.005	0.892 ± 0.004	0.041 ± 0.004	0.830 ± 0.012	0.927 ± 0.003	6.4 ± 0.5	0.2 ± 0.1	—
C2 linear N100	0.932 ± 0.003	0.051 ± 0.008	0.892 ± 0.005	0.036 ± 0.003	0.826 ± 0.012	0.926 ± 0.002	6.4 ± 0.6	0.2 ± 0.1	—
C3 linear N200	0.925 ± 0.007	0.051 ± 0.006	0.893 ± 0.004	0.031 ± 0.004	0.819 ± 0.010	0.920 ± 0.006	6.2 ± 0.6	0.1 ± 0.0	—
C4 adaptive	0.917 ± 0.001	0.019 ± 0.004	0.931 ± 0.005	0.016 ± 0.003	0.798 ± 0.010	0.915 ± 0.001	2.5 ± 0.5	0.3 ± 0.1	—
C5 adaptive lmin0.05	0.922 ± 0.001	0.023 ± 0.004	0.930 ± 0.005	0.016 ± 0.002	0.802 ± 0.009	0.921 ± 0.001	2.6 ± 0.5	0.1 ± 0.0	—
C6 adaptive lmin0.10	0.927 ± 0.002	0.028 ± 0.005	0.926 ± 0.004	0.017 ± 0.003	0.806 ± 0.009	0.927 ± 0.002	3.0 ± 0.4	0.0 ± 0.0	—
C7 adaptive lmin0.20	0.931 ± 0.002	0.035 ± 0.005	0.919 ± 0.003	0.021 ± 0.004	0.812 ± 0.009	0.930 ± 0.001	3.7 ± 0.4	0.0 ± 0.0	—

Table C.2: rot-MNIST λ-curriculum sweep, full metrics, at both momentum settings, each block headed by the vanilla-ER reference the sweep is quoted against. All conditions applied on top of standard ER (1 k reservoir). Area is on the step axis, as in every block at fixed η; the ladder converts to flow time (Table C.1).

The ingredient sweep: adaptive signal and $\lambda_{\min}$ floor. Table C.2 settles the two design choices of the shipped configuration, and both are decided in the momentum-on leg. At momentum 0 the adaptive schedule is indistinguishable from the longest linear ramp (12.8 pp / 2.2 / 0.833 against C3's 14.3 pp / 1.8 / 0.828, seeds split on depth, $p = 0.44$), so at that setting it is preferred for having no ramp length to tune rather than for its numbers. Under momentum the same pair separates: $C4_M$ reaches 2.5 pp / 0.3 / 0.917 against $C3_M$'s 6.2 pp / 0.1 / 0.925, shallower in all five seeds by $\approx$3.7 pp, so the feedback signal both removes the knob and beats the best setting of the knob, at a small cost in final accuracy and in the tail. The floor trades that cost back. Raising $\lambda_{\min}$ from 0 to 0.20 ($C4_M \to C7_M$) moves depth from 2.5 to 3.7 pp, worse in all five seeds, while accuracy moves from 0.917 to 0.931 and the area from 0.3 to 0.03, better in all five seeds on both; the intermediate settings $C5_M$ and $C6_M$ walk monotonically between the two ends in depth and accuracy, so the floor is a graded Pareto knob rather than a switch. $C7_M$ is the shipped point because it clears the tuned linear ramp on all three quantities (depth in five seeds of five, accuracy in four) and returns accuracy to vanilla ER's level (3.7 pp / 0.03 / 0.931 against 15.9 pp / 1.0 / 0.928, above vanilla in four of five seeds and within the seed spread). At momentum 0 the floor is not a useful knob: it costs depth (12.8 $\to$ 13.9 pp) and returns almost nothing in accuracy (0.833 $\to$ 0.837), since early T_1 progress there is set by the boundary gradient the floor is too small to redirect.

The sweep as trajectories. Figures C.1 and C.2 are the trajectory form of Table C.2, one figure per schedule family on the same 2×2: rows the momentum setting, columns stability (T_0) and plasticity (T_1), vanilla ER carried into every panel as the ungated reference. Read together they show why the two knobs are decided in the momentum-on leg. At momentum 0 both families trade the same way and neither trade is good: the linear ramps buy their shallower start by postponing the descent rather than preventing it, each rung releasing the brake at its own N and dropping into the same chronic, jittery band vanilla ER occupies (Figure C.1(a)), and every rung of both families pays for it in (b), where the new-task curve is displaced bodily to the right in proportion to how much schedule it carries. That displacement is the plasticity debt, and at momentum 0 nothing repays it. The momentum row is a different picture. In (c) the schedules never dip at all: they descend gently from the pre-switch level and stay above vanilla ER for essentially the whole transition instead of dropping below it and recovering, and in (d) the new-task curves have closed most of the way back onto vanilla's, the debt repaid, with the residual gap ordered by the floor: the more the schedule is allowed to brake, the more of the debt survives acceleration, which is what makes $\lambda_{\min}$ a graded Pareto knob rather than a switch. Both stability panels of a figure span the same 0.30 of accuracy on an offset floor, so a deeper-looking dip is a deeper dip.

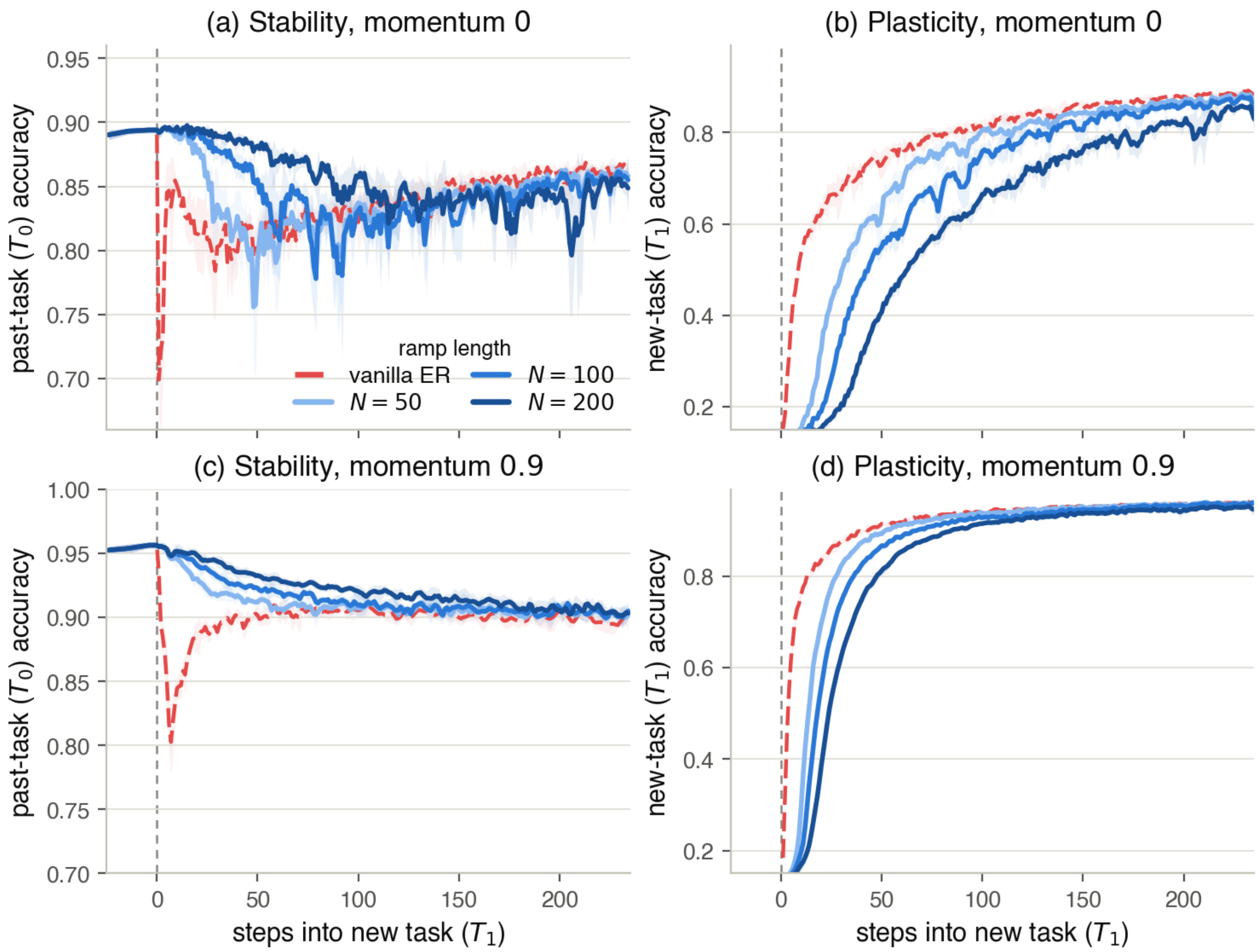


Figure C.1: The linear λ-ramps (C1–C3) across the rot-MNIST switch: per-step accuracy, seed mean with $\pm$std bands, x=0 the task switch. Rows are the momentum setting, columns the two tasks; the ramp runs short light $\rightarrow$ long dark, over vanilla ER (red, dashed). A longer ramp holds T_0 higher for longer and costs correspondingly more early T_1 progress, at momentum 0 without ever escaping the chronic band vanilla settles into.

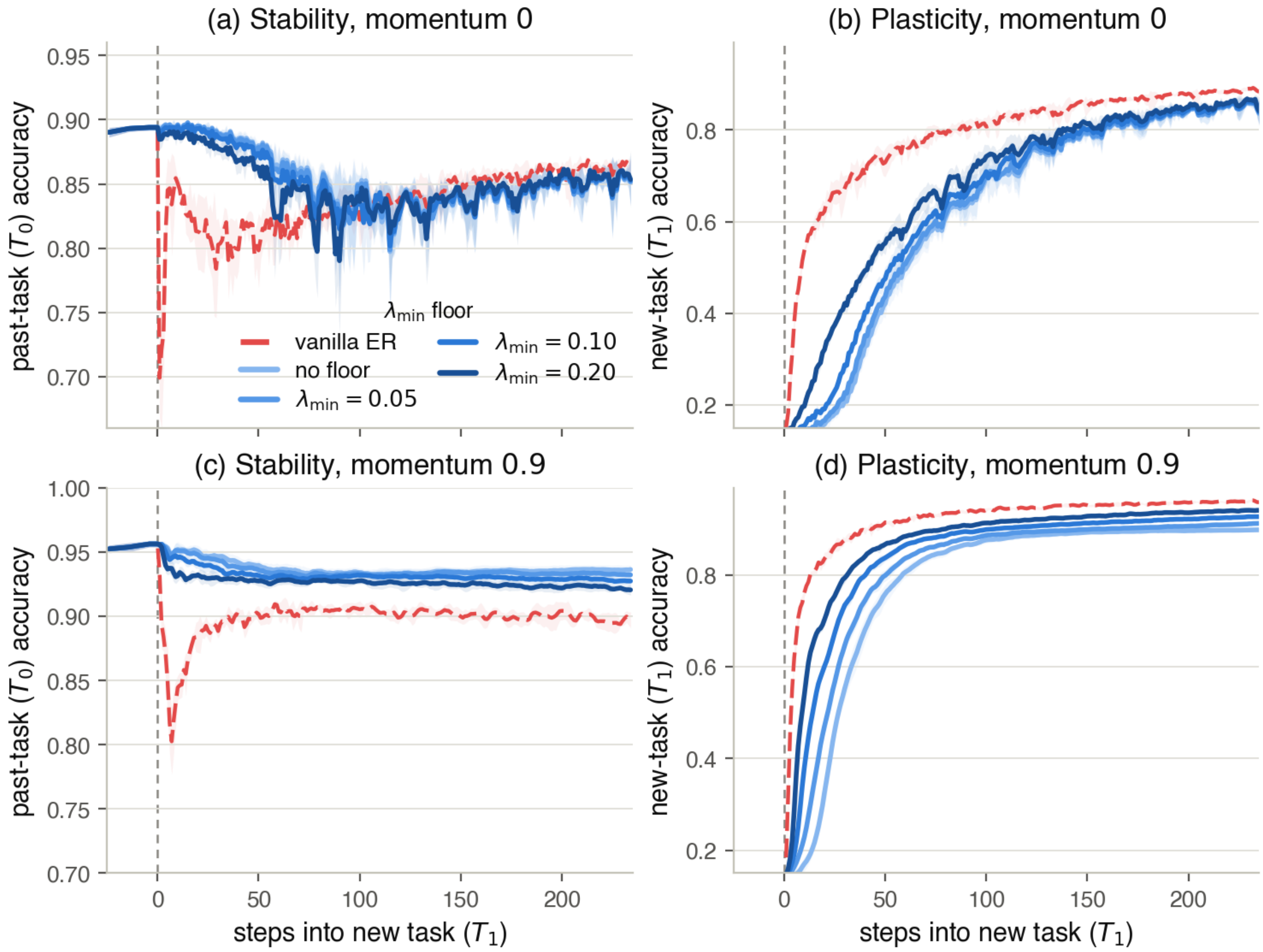


Figure C.2: The adaptive λ-curriculum and its $\lambda_{\min}$ floor (C4–C7), drawn as Figure C.1; the ramp runs unfloored light → $\lambda_{\min}$=0.20 dark. Raising the floor returns early T_1 progress (panels (b) and (d), the darker rungs leading) for a slightly deeper T_0 transient, the trade the shipped configuration C7 takes at momentum 0.9 (Table C.2).

Full-buffer momentum diagnostic (C8). Table C.3 crosses the practical curriculum with the full 60k buffer at both momentum settings, isolating momentum's two roles (Section 6.2). Source variance reduction alone (C8, full buffer, μ=0) already closes the gap to 0.2 pp (Figure C.3), so the residual the curriculum leaves at μ=0 is estimator variance, not structure; momentum's gap-closing role is interchangeable with the full buffer. Acceleration alone (the μ=0.9 column) restores accuracy: C8$_M$ reaches 0.955, approaching the exact-gradient ceiling (the ladder's ungated $\eta = 0.1$ rung on the same 60k gradient at μ=0.9: 0.965, Table C.1), while re-introducing a small transient (2.0 pp, area 0.38). The shape and timing of this transient point to the $\lambda_{\min}$ floor meeting momentum. With the replay gradient exact and the arc removed, the only push left in the first steps is the floored fifth of the new-task gradient (the logged gradient-norm ratio confirms the schedule sits at $\lambda = 0.2$ through the dip window), and momentum amplifies a persistent push toward $1/(1-\mu) = 10$ times its per-step size: the dip accordingly opens at the boundary, bottoms out about ten steps in, and closes within about fifty as the replay gradient reasserts itself. The floor is by construction the one share of the push the source gate does not attenuate, so momentum briefly re-inflates exactly that share, a miniature of the directional gate's re-inflation. The dose-response on the same exact buffer supports the reading: the floored push without momentum costs 0.2 pp (C8), with momentum 2.0 pp (C8$_M$), and the unfloored abrupt switch with momentum 14.1 pp (the same ungated exact-gradient rung, which is not seed-paired with C8$_M$ and is read as a magnitude only). C8$_M$'s area additionally counts in full under the end-referencing convention of Section 5.1.2, since it recovers above its pre-switch level. C8/C8$_M$ are diagnostic probes: the full buffer defeats the limited-memory premise, and the practical scalar-gate configuration remains C7$_M$ (3.7 pp / 0.03 / 0.931) at 1/60th the memory.

	C7 (1k, μ=0)	C7$_M$ (1k, μ=0.9)	C8 (full, μ=0)	C8$_M$ (full, μ=0.9)
`gap_depth` (pp)	13.9	3.7	**0.2**	2.0
`gap_area_end`	2.7	0.03	**0.05**	0.38
ACC	0.837	0.931	0.818	**0.955**

Table C.3: Curriculum $\times$ full buffer $\times$ momentum on rot-MNIST, isolating momentum's two roles. Source variance reduction alone (C8) closes the gap; acceleration alone (the μ column) restores accuracy. The residual at C7 (μ=0) is variance, not structure.

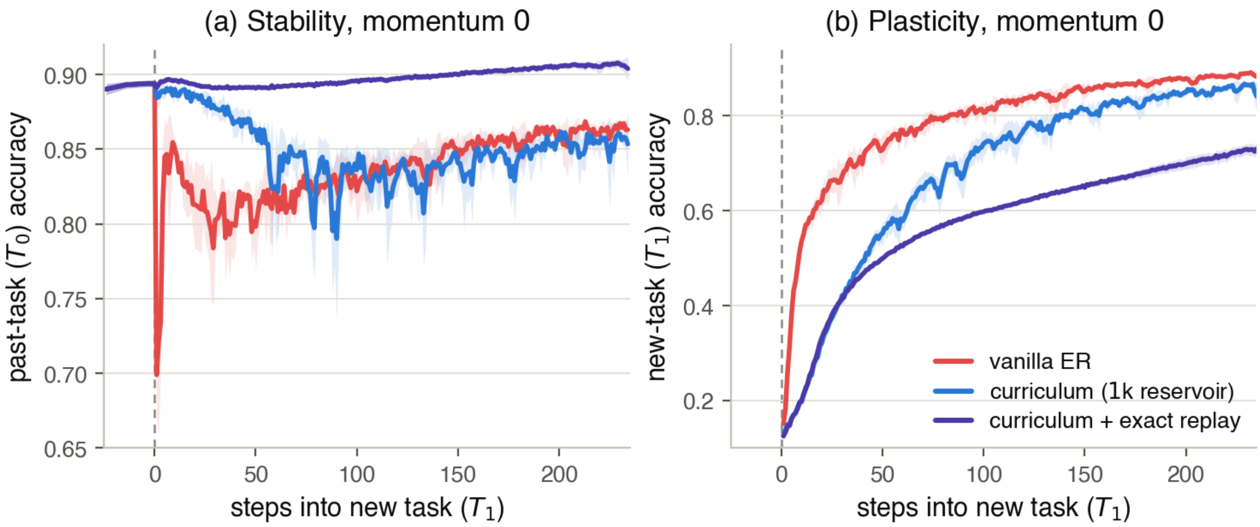


Figure C.3: The estimator-variance control at momentum 0: per-step accuracy across the rot-MNIST switch, seed mean with $\pm$std bands, for vanilla ER (red), the practical curriculum C7 on the 1k reservoir (blue), and the same schedule on the exact 60k replay gradient C8 (violet). (a) Stability (T_0): C7 removes vanilla's boundary spike but keeps a jittery chronic drop that bottoms out near 0.79; with the replay gradient exact, the same schedule never leaves its pre-switch plateau (0.2 pp of depth against C7's 13.9, Table C.3). The seed band reads the same way as the mean, widening to nearly 6 pp across C7's transient and staying under one for C8. What C7 still leaks at this momentum setting is therefore the noise of the limited buffer, not a second deformation of the path. (b) Plasticity (T_1): what the control costs. Defending the past task at every step makes new-task learning the slowest of the three, and with no momentum to repay the debt the epoch ends some 11 pp behind C7, which is why C8 posts the lowest ACC of the four cells in Table C.3 while holding the smallest gap.

The gate against the ladder's bar, rung by rung. Lowering the gap is not by itself the gate's claim, because the step size lowers it with no method at all; the claim has to be that the gate lowers it at the standard step size and in the standard number of steps (Section 6.1). Nor is there an accuracy handicap to discount from the ladder's numbers: ACC is flat to rising down both arms and min-ACC improves sharply, so what the ladder charges is compute, the epoch count scaling as $1/\eta$ at matched flow time. Read against that bar on the matched 1k arm at momentum 0, the curriculum loses on everything but the budget. C7 leaves 13.9 pp of depth at ACC 0.837 and min-ACC 0.743 in one epoch at $\eta = 0.1$; the $\eta = 0.01$ rung leaves 4.9 pp at ACC 0.877 and min-ACC 0.832 in ten. Measured against its own protocol's reference the schedule removes 5.6 pp of vanilla ER's 19.5 and spends 3.6 pp of ACC doing it ($0.873 \to 0.837$), where the tenfold shorter step removes 18.0 pp of the ladder's distributionally matched 22.9 and gains 0.3 pp of ACC ($0.874 \to 0.877$). The flow-time areas do land together, 0.27 for C7 against the $\eta = 0.01$ rung's 0.275 (the curriculum's step-axis 2.7 converted at $A_\tau = \eta A_{\text{steps}}$), but the two are referenced to plateaus four points apart and a lower plateau is a lower line to integrate against, so this is not a tie (Section 5.1.2).

Under momentum the comparison turns around, which is the reading Section 6.2 quotes. On the same 1k arm at momentum 0.9 the $\eta = 0.01$ rung reaches 5.4 pp at ACC 0.940 and min-ACC 0.901 after ten epochs, so one epoch of the schedule is 1.7 pp shallower and 1.8 pp better in worst-case retention than ten epochs of a tenfold shorter step, for 0.9 pp of ACC (the two protocols' $\eta = 0.1$ cells agree here as at momentum 0, 15.4 ± 2.2 against 15.9 ± 2.6 pp at ACC 0.929 against 0.928). That rung is a rung and not a limit, but the ladder's fourth rung supplies the limit as well: at $\eta = 0.001$, a hundred epochs, the same arm reaches 4.6 pp at ACC 0.942 and min-ACC 0.910, and the fine-rung extrapolation puts the $\eta \to 0$ depth of this leg at 4.1 pp. One epoch of the schedule is thus 0.9 pp shallower than a hundred epochs of a hundredfold shorter step, and its 3.7 ± 0.4 pp sits at or below what shortening the step can ultimately reach here, for 1.1 pp of ACC.

C.3 rot-MNIST: Buffer-Fidelity Diagnostics

The gradient diagnostics of Appendix B were logged on the vanilla-ER reference runs only. Each such step costs a full forward and backward pass over the whole past-task training set, which is why they are not carried elsewhere, and the stretched rungs of the ladder least of all (Appendix A.4). The rows below are therefore the directional and magnitude error that a 1,000-sample reservoir incurs against g_{true} at the standard step size, and they set the figure the ladder's exact arm is defined against: a 60k buffer replays the deterministic empirical past-task gradient, so its cosine and magnitude ratio are 1.000 by construction.

Condition	$\bar{\cos}$	$\cos_{\min}$	$\bar{\rho}$
D1 vanilla ER (μ=0)	0.711 ± 0.035	0.086 ± 0.103	1.040 ± 0.039
D1 vanilla ER (μ=0.9)	0.551 ± 0.033	0.135 ± 0.069	0.418 ± 0.014

Table C.4: Buffer-fidelity diagnostics for the vanilla-ER reference: $\bar{\cos}$ and $\cos_{\min}$ are the mean and minimum of $\cos(g_{\text{replay}}, g_{\text{true}})$ over the window; $\bar{\rho}$ is the mean of $\|g_{\text{replay}}\|/\|g_{\text{true}}\|$.

C.4 rot-MNIST: Asymmetric PER (P-series)

Table C.5 gives the full metric set for the asymmetric-PER δ sweep on both buffer legs and at both momentum settings; the δ=0.03 rows are the two the main text carries (Table 6.1), and the momentum-0.9 rows are the source of the re-inflation numbers of Section 6.3. Momentum-0.9 runs cover the three strongest filters on the standard leg and $\delta \in \{0.3, 0.1\}$ on the full-buffer leg.

Condition	ACC	FORG	min-ACC	WF_{100}	WP_{100}	WC-ACC	Depth	Area end
Standard (1k) buffer, momentum 0.0								
δ=1.0	0.869 ± 0.005	0.027 ± 0.018	0.792 ± 0.012	0.081 ± 0.009	0.733 ± 0.009	0.838 ± 0.006	9.0 ± 0.6	3.29 ± 0.65
δ=0.3	0.873 ± 0.004	0.023 ± 0.017	0.807 ± 0.012	0.068 ± 0.009	0.736 ± 0.008	0.847 ± 0.005	7.4 ± 1.1	2.32 ± 0.59
δ=0.1	0.876 ± 0.004	0.018 ± 0.016	0.822 ± 0.012	0.056 ± 0.009	0.739 ± 0.008	0.856 ± 0.005	5.9 ± 1.5	1.56 ± 0.65
δ=0.03	0.880 ± 0.005	0.012 ± 0.015	0.834 ± 0.012	0.048 ± 0.005	0.741 ± 0.007	0.862 ± 0.005	4.8 ± 1.9	1.04 ± 0.58
Standard (1k) buffer, momentum 0.9								
δ=0.3	0.929 ± 0.002	0.061 ± 0.004	0.851 ± 0.008	0.065 ± 0.004	0.813 ± 0.009	0.907 ± 0.005	10.4 ± 0.5	0.50 ± 0.17
δ=0.1	0.928 ± 0.002	0.061 ± 0.006	0.861 ± 0.008	0.060 ± 0.005	0.815 ± 0.009	0.911 ± 0.005	9.4 ± 0.5	0.38 ± 0.19
δ=0.03	0.929 ± 0.003	0.060 ± 0.009	0.871 ± 0.008	0.053 ± 0.007	0.816 ± 0.010	0.917 ± 0.005	8.5 ± 0.6	0.23 ± 0.12
Full (60k) buffer, momentum 0.0								
δ=1.0	0.898 ± 0.002	-0.025 ± 0.012	0.841 ± 0.011	0.037 ± 0.011	0.734 ± 0.006	0.865 ± 0.006	4.0 ± 1.0	2.32 ± 1.38
δ=0.3	0.900 ± 0.002	-0.029 ± 0.012	0.867 ± 0.002	0.026 ± 0.007	0.739 ± 0.005	0.878 ± 0.002	1.5 ± 1.1	1.01 ± 0.91
δ=0.1	0.901 ± 0.002	-0.032 ± 0.013	0.860 ± 0.026	0.040 ± 0.028	0.741 ± 0.004	0.875 ± 0.013	2.2 ± 1.6	0.44 ± 0.38
δ=0.03	0.903 ± 0.002	-0.035 ± 0.012	0.787 ± 0.094	0.118 ± 0.102	0.742 ± 0.004	0.838 ± 0.048	9.5 ± 8.2	0.43 ± 0.25
δ=0.03 (CG 60)	0.903 ± 0.002	-0.035 ± 0.012	0.766 ± 0.113	0.126 ± 0.109	0.742 ± 0.004	0.828 ± 0.057	11.5 ± 10.1	0.47 ± 0.28
Full (60k) buffer, momentum 0.9								
δ=0.3	0.964 ± 0.001	-0.012 ± 0.002	0.864 ± 0.009	0.052 ± 0.005	0.813 ± 0.008	0.913 ± 0.005	9.1 ± 0.8	1.91 ± 0.26
δ=0.1	0.965 ± 0.001	-0.012 ± 0.003	0.880 ± 0.008	0.044 ± 0.004	0.814 ± 0.008	0.921 ± 0.005	7.6 ± 0.7	1.57 ± 0.25

Table C.5: rot-MNIST asymmetric-PER δ sweep, full metrics: both buffer legs at both momentum settings, plus the solver control (CG 60). The momentum-0.9 rows show the re-inflation of Section 6.3: depth climbs on every leg relative to its momentum-0 counterpart, while the full-buffer accuracies ($\approx$0.965) are diagnostic values only, the full buffer defeating the limited-memory premise.

The sweep, rung by rung, on both tasks. Figure C.4 is the trajectory form of Table C.5: the whole standard-buffer sweep, both tasks, both momentum settings. Three things are visible there that the scalar summaries only assert. First, the ordering in δ in (a) is an ordering of whole trajectories and not only of their worst points: the rungs stack without crossing for the entire transition, so a stronger filter is uniformly better on the past task rather than better at the spike and worse afterwards. Second, (b) is the evidence behind the no-plasticity-cost claim of Section 6.3: the strongest filter is visibly slower over roughly the first fifty steps, the selective slowdown along the directions the sampled Fisher rates as sharp, and thereafter it is indistinguishable from vanilla ER, so the horizon-100 metrics record no cost. Third, the momentum row keeps that ordering in (c) but reads it against a baseline that has itself changed: vanilla ER's transient is shorter and the whole ladder sits 3–4 pp deeper, while (d) shows the new-task curves collapsing onto one another, momentum having removed what little separated them. Both stability panels span the same 0.30 of accuracy on an offset floor, so a deeper-looking dip is a deeper dip; the pre-switch level is higher in (c) only because momentum converges T_0 further before the switch. No δ=1.0 cell was run at momentum 0.9.

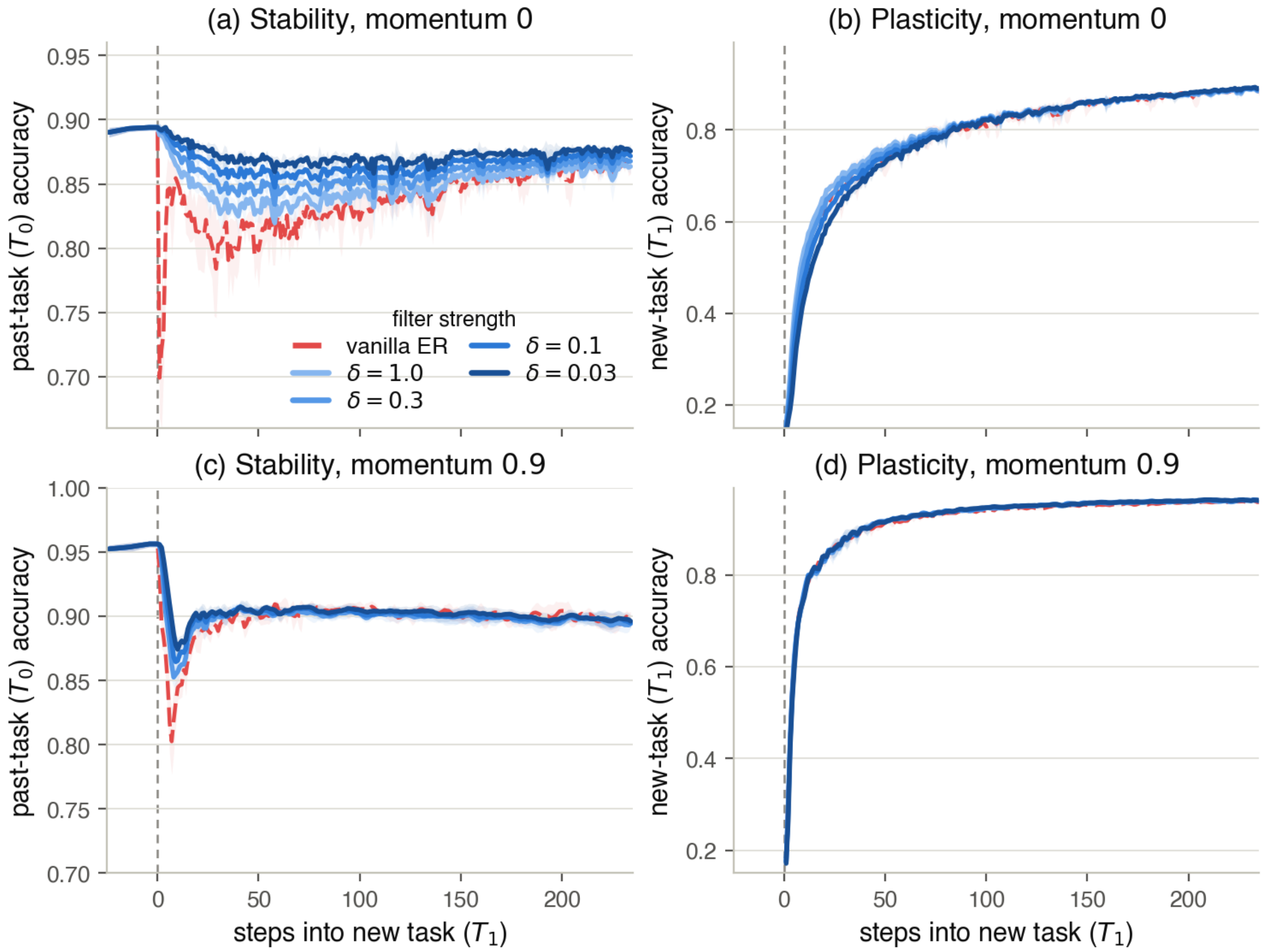


Figure C.4: The asymmetric-PER δ sweep on the standard (1k) buffer: per-step accuracy across the rot-MNIST switch, seed mean with $\pm$std bands, x=0 the task switch. Rows are the momentum setting, columns the two tasks; the damping runs weaker filter light $\rightarrow$ stronger filter dark, over vanilla ER (red, dashed) as the ungated reference in every panel. (a) the dip shrinks monotonically as $\delta \rightarrow 0$; (b) the strongest filter lags for $\sim$50 steps and then tracks vanilla; (c) the same ordering, displaced 3–4 pp deeper by the accumulator; (d) under momentum the new-task curves are indistinguishable. Scalar metrics for every rung are in Table C.5.

The exact-gradient diagnostic and its cliffs. The full (60k) buffer is a diagnostic probe rather than an operating point: replaying the exact past-task gradient switches the variance driver off, so whatever excursion survives is the arc together with the standard step's overshoot of it (Section 6.1). Read that way the leg behaves as the account predicts down to $\delta = 0.1$, the excursion shrinking with the filter. The one place the sweep breaks monotonicity is the strongest setting, $\delta = 0.03$, where mean depth jumps to 9.5 pp with a std of 8.2: in 3 of 5 seeds the past-task curve holds flat for roughly two hundred steps and then slides off a late cliff, the red per-seed traces in Figure C.5. A solver control rules out the most straightforward explanation. The filter solves for its preconditioned direction by conjugate gradient, so an under-converged solve at the strongest damping would be the obvious suspect; rerunning the setting with the CG solver at 60 iterations instead of the default 25 (row δ=0.03 *(CG 60)* of Table C.5) reproduces the cliffs at the same seeds and the same steps, and if anything deepens them (11.5 ± 10.1 pp mean depth against 9.5 ± 8.2). They are not a numerical artefact of the solve. The cliffs came as a surprise: the account made no prediction of a late, abrupt failure, and nothing in the rest of the sweep hints at one. They are also confined to the exact-gradient diagnostic: at the same δ on the standard 1k buffer no seed shows one (depth 4.8 ± 1.9 pp, the monotone end of the sweep). The failure is tied to the exact replay gradient: the sampling jitter of a practical buffer appears to protect against it, the same noise that elsewhere drives the overshoot here keeping the iterate from settling into the escape direction. Why the cliffs happen we can only partly say. A reading consistent with the data is a blind spot of the gate's model of the past task: the filter passes new-task pressure through directions its sampled Fisher rates as flat, and in these seeds that trust appears to be misplaced further out along the trajectory. We offer this as a hypothesis, not an established mechanism. What the control does establish is that the strongest filter can fail late and abruptly when nothing perturbs it. Because the failure is confined to the diagnostic leg, it is unlikely in a practical configuration, but it exposes a real weakness of feedforward control.

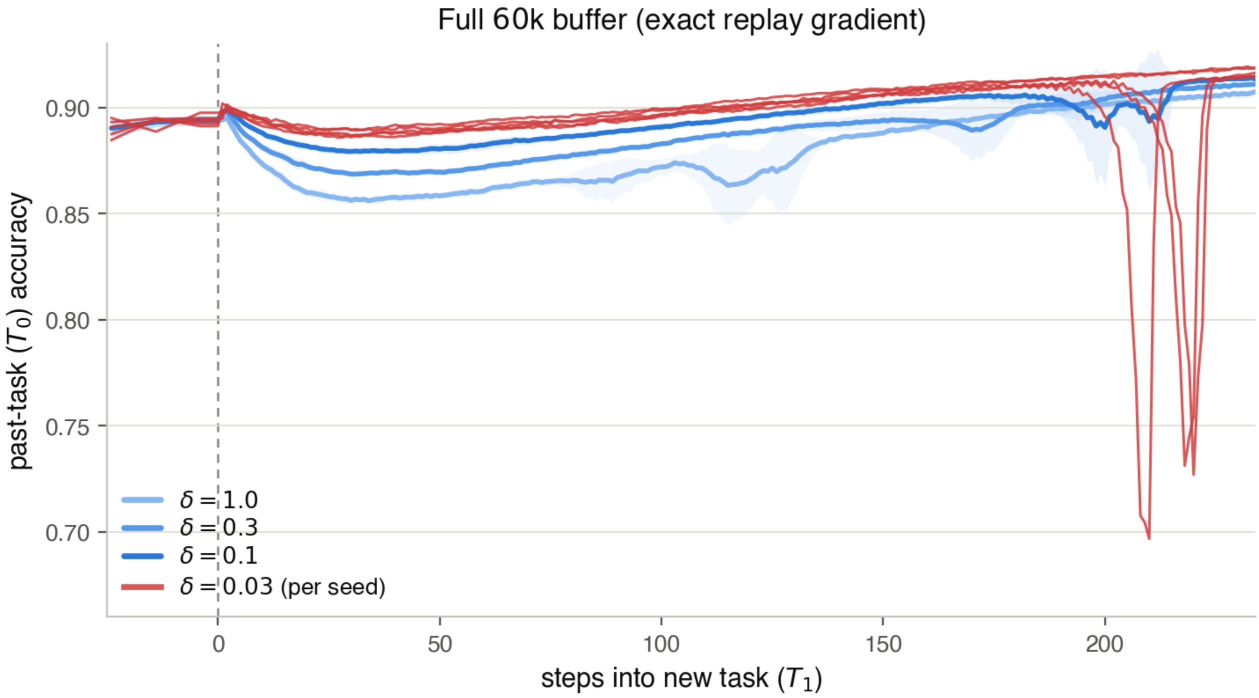


Figure C.5: The exact-gradient diagnostic leg of the δ sweep: per-step past-task (T_0) accuracy across the rot-MNIST switch at momentum 0 on the full (60k) buffer, where the variance driver is off and the residual excursion is the arc together with the standard step's overshoot of it. The excursion shrinks with the filter for $\delta \geq 0.1$ (seed mean with ±std bands, weaker filter light → stronger filter dark), but at the strongest setting δ=0.03 the gate fails late and abruptly: shown per seed (critical red), 3 of 5 seeds hold flat for ∼200 steps and then slide off a cliff. The failure is confined to this diagnostic leg; at the same δ on the standard 1k buffer no seed shows one (Figure C.4(a), depth 4.8 ± 1.9 pp).

C.5 rot-MNIST: Symmetric vs. Asymmetric Preconditioning

Leaving the restoring gradient g_{rep} at full strength is the directional gate's defining choice (Section 6.3). The alternative is to precondition the summed update, $d = \delta(F_{\text{rep}} + \delta I)^{-1}(g_{\text{cur}} + g_{\text{rep}})$, the symmetric variant that passes both gradients through the same metric. Because g_{rep} is the gradient of the replay loss, it lies in the top eigenspace of that loss's own Fisher F_{rep}, so the symmetric filter shrinks the restoring gradient *harder* than the harmful push it is meant to brake and cannot preferentially protect the past task. Table C.6 sweeps the same δ on both variants, and they are mirror images. As the filter strengthens ($\delta \to 0$) the asymmetric gate's forgetting falls (FORG 0.027 → 0.012) and its accuracy rises, while the symmetric gate's forgetting *climbs* (0.030 → 0.066) and its accuracy falls; every one of these four trends holds across all five seeds. Symmetric depth barely moves across the sweep (5.7 → 6.7 pp) and at δ=1.0 even sits below the asymmetric gate's (5.7 vs. 9.0 pp), because filtering the whole update also damps the per-step *expression* of the dip. This is the area caveat of Section 5.1.2 made concrete: the symmetric gate's end-referenced area collapses toward zero (2.7 → 0.01) not because the dip recovers but because it stops recovering, settling into a permanent accommodation that the area excludes and the forgetting metric records. Read on depth or transient area the symmetric gate looks competitive; read on forgetting and final accuracy it degrades monotonically as its filter strengthens. Asymmetry is what turns the filter from a reshuffling of the damage into a net removal of it.

	Asymmetric (g_{rep} free)				Symmetric (joint)			
δ	depth (pp)	area	FORG	ACC	depth (pp)	area	FORG	ACC
1.0	9.0	3.29	0.027	0.869	5.7	2.69	0.030	0.872
0.3	7.4	2.32	0.023	0.873	5.9	1.65	0.042	0.867
0.1	5.9	1.56	0.018	0.876	6.2	0.34	0.057	0.861
0.03	4.8	1.04	0.012	0.880	6.7	0.01	0.066	0.857

Table C.6: Asymmetric versus symmetric preconditioning on the standard 1k buffer at momentum 0, five seeds, sweeping the shared damping δ. Both variants filter through the replay Fisher F_{rep}; they differ only in whether the metric is applied to the new-task push alone (asymmetric, g_{rep} left free) or to the summed update (symmetric). As $\delta \to 0$ the asymmetric gate improves on every axis while the symmetric gate degrades on every axis; symmetric depth stays flat because the filter suppresses the per-step expression of the dip even as the underlying forgetting grows, and the symmetric end-referenced area collapses toward zero for the same reason: the past task stops recovering, so the loss registers in FORG rather than in the transient area (Section 5.1.2).

C.6 CIFAR-10: Headline Block

Condition	ACC	FORG	min-ACC	WF_{100}	WP_{100}	WC-ACC	Depth	Area end
G1 vanilla BN	0.723 ± 0.008	0.010 ± 0.019	0.644 ± 0.023	0.127 ± 0.015	0.484 ± 0.020	0.667 ± 0.016	11.8 ± 2.5	17.5 ± 4.6
G2 adaptive BN	0.722 ± 0.011	0.006 ± 0.016	0.713 ± 0.017	0.070 ± 0.010	0.469 ± 0.016	0.698 ± 0.013	4.8 ± 0.7	3.6 ± 2.2
G3 asym. PER BN	0.715 ± 0.011	-0.008 ± 0.018	0.698 ± 0.023	0.124 ± 0.022	0.460 ± 0.025	0.682 ± 0.023	5.9 ± 4.3	3.3 ± 3.0
G4 vanilla GN	0.730 ± 0.011	-0.023 ± 0.030	0.309 ± 0.111	0.294 ± 0.097	0.415 ± 0.058	0.507 ± 0.054	41.2 ± 13.5	34.9 ± 23.7
G5 adaptive GN	0.731 ± 0.020	-0.026 ± 0.026	0.670 ± 0.055	0.095 ± 0.033	0.340 ± 0.052	0.687 ± 0.037	5.2 ± 2.1	1.7 ± 0.9
G6 asym. PER GN	0.741 ± 0.009	-0.035 ± 0.018	0.673 ± 0.023	0.080 ± 0.009	0.330 ± 0.024	0.696 ± 0.015	5.6 ± 1.1	4.8 ± 3.8

Table C.7: CIFAR-10 headline block, full metrics (ResNet-18, momentum 0.9, buffer 5,000, five seeds). "adaptive" is the practical curriculum with $\lambda_{\min} = 0.20$; "asym. PER" runs at $\delta = 0.1$.

The gates' residual batch-norm dip. The small gap the gates keep under batch norm is a genuine boundary transient: in Figure 6.4(a) both gated curves dip together over the first steps after the switch, by roughly the tabulated 5 pp, and recover to their plateau within a few tens of steps. A reading consistent with this pattern is the normalisation pathway of Section 5.2.5. BatchNorm's running statistics update on every forward pass, whatever the gate does to the gradient, so the first corrupted batches shift them away from the clean task and no gate on the update can hold that back; the clean replay samples in every batch pull the statistics back, which matches the fast recovery. The group-norm contrast supports the reading: with no cross-batch statistics to disturb, the curriculum holds T_0 flat through the same boundary (Figure 6.4(b)) and its transient area falls to roughly half its batch-norm value (1.7 vs. 3.6). This is a reading rather than an established mechanism; a control that freezes the running statistics across the switch would isolate it.

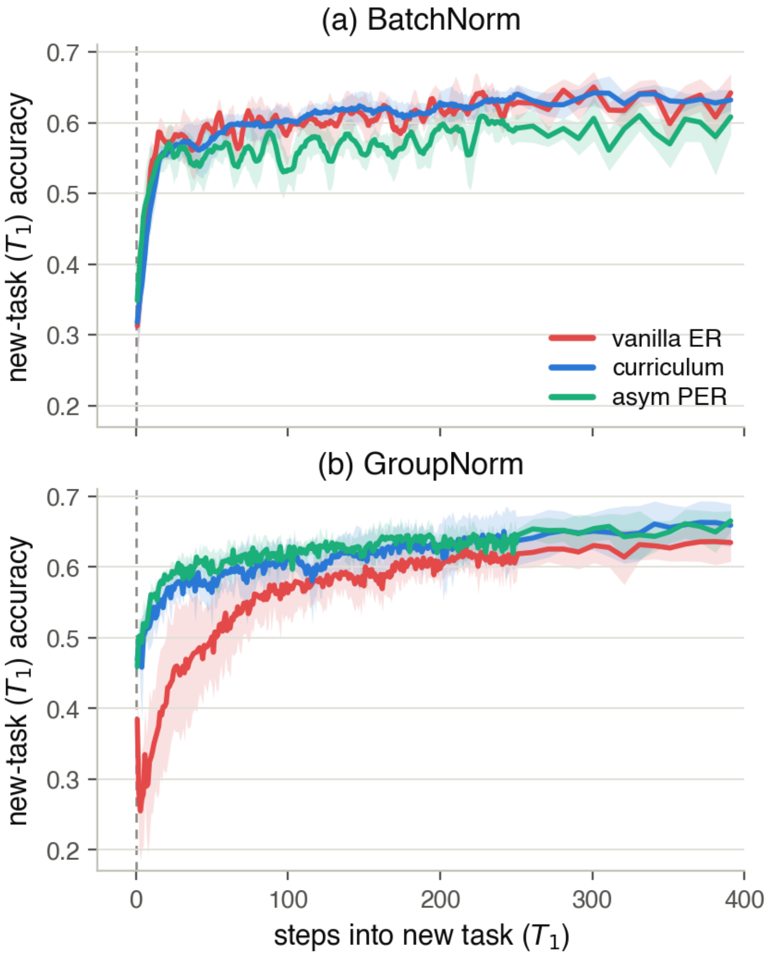


Figure C.6: What the deep-backbone protection costs on the new task: the plasticity counterpart of Figure 6.4, over the same conditions and the same post-switch window, but tracking new-task (T_1) accuracy across the clean → Gaussian-noise switch (ResNet-18, momentum 0.9, buffer 5,000, seed mean with ±std bands). (a) BatchNorm: both gates track vanilla ER, the directional gate lagging it by a few points over the first hundred steps (0.53 against 0.59 at step 100) before closing; after ten epochs the three land within 2.5 pp of each other (0.69, 0.68, 0.67). (b) GroupNorm: the gates are *faster* than vanilla ER on the new task at step 50, by 6 pp (scalar gate) and 11 pp (directional gate), and the directional gate finishes ahead of it (0.72 against 0.70). Where the ungated baseline loses the past task it also loses the shared features the new task starts from, so on this backbone the protection is not bought from plasticity, which is what keeps the ACC column of Table C.7 flat across the batch-norm rows and rising across the group-norm ones.

C.7 CIFAR-10: Generalisation Block

Each shift pairs the practical curriculum ("curric.": adaptive $\lambda_{\min} = 0.20$, buffer 5,000) against a vanilla-ER control on the same shift, at the indicated normalisation, over three seeds.

Condition	ACC	FORG	min-ACC	WC-ACC	Depth	Area end
BatchNorm						
shot noise (vanilla)	0.724 ± 0.011	0.017 ± 0.010	0.633 ± 0.017	0.663 ± 0.014	13.1 ± 3.0	21.2 ± 3.9
shot noise (curric.)	0.723 ± 0.014	0.010 ± 0.010	0.708 ± 0.032	0.697 ± 0.016	5.6 ± 1.0	5.8 ± 1.5
contrast (vanilla)	0.671 ± 0.026	-0.008 ± 0.006	0.610 ± 0.063	0.586 ± 0.048	15.4 ± 3.9	33.4 ± 4.5
contrast (curric.)	0.671 ± 0.044	-0.012 ± 0.007	0.727 ± 0.034	0.643 ± 0.048	3.7 ± 1.1	2.2 ± 0.9
3-task (vanilla)	0.740 ± 0.006	-0.011 ± 0.013	0.670 ± 0.015	0.691 ± 0.010	3.3 ± 0.4	6.0 ± 0.7
3-task (curric.)	0.745 ± 0.010	-0.021 ± 0.008	0.677 ± 0.025	0.696 ± 0.017	6.9 ± 2.7	3.9 ± 0.1
GroupNorm						
shot noise (vanilla)	0.718 ± 0.021	-0.036 ± 0.036	0.244 ± 0.075	0.469 ± 0.032	46.3 ± 12.0	54.4 ± 40.8
shot noise (curric.)	0.705 ± 0.061	-0.028 ± 0.012	0.638 ± 0.108	0.656 ± 0.087	7.3 ± 5.9	2.4 ± 1.1
contrast (vanilla)	0.777 ± 0.022	-0.067 ± 0.021	0.641 ± 0.061	0.710 ± 0.040	8.4 ± 4.4	3.3 ± 0.6
contrast (curric.)	0.778 ± 0.032	-0.070 ± 0.013	0.668 ± 0.077	0.724 ± 0.054	3.9 ± 3.1	1.2 ± 0.8
3-task (vanilla)	0.722 ± 0.020	-0.023 ± 0.015	0.468 ± 0.043	0.550 ± 0.023	4.4 ± 0.4	10.3 ± 3.6
3-task (curric.)	0.721 ± 0.019	-0.025 ± 0.020	0.421 ± 0.114	0.517 ± 0.070	32.8 ± 14.5	32.0 ± 19.5

Table C.8: CIFAR-10 generalisation block, full metrics. The two-task novel corruptions (shot noise, contrast) favour the curriculum under both normalisations; the three-task group-norm sequence is the outlier analysed below.

The three-task group-norm failure. The three-task sequence is the one place the curriculum underperforms its control (depth 32.8±14.5 pp against vanilla's 4.4±0.4, Table C.8; Figure C.7), and the logged gradient-norm traces locate the failure in the conditioning of the feedback signal. The adaptive ratio $r = \|g_{\text{replay}}\|/\|g_{\text{new}}\|$ carries information through the boundary ordering of Section 5.2.2: at a well-separated switch the new-task gradient dwarfs the replay gradient, so r starts near zero and its rise tracks the replay gradient reasserting itself. The first boundary of this sequence behaves that way (clean → Gaussian: first-step new-task loss 1.9–2.1, $r \approx 0.2$), and the schedule holds every past task flat through it. The second boundary never establishes the ordering: Gaussian and shot noise are near-identical, so the incoming model already fits the new task (T_2 accuracy 0.67–0.71 before its first step, new-task loss 0.5–0.7), leaving $\|g_{\text{new}}\|$ small, while $\|g_{\text{replay}}\|$ is small because the model is converged on the replayed tasks. The ratio becomes a quotient of two near-zero norms: under group norm it reads 0.5–1.3 from the first step and fluctuates step to step, so the EMA drives λ from the floor to ≈0.7–0.8 within fifty steps and back to $\lambda_{\min}$ within a few hundred. The gate is moved by noise, not by damage.

What follows, in all three seeds, is not a stability gap but an optimisation instability. Within the first forty steps the replay loss itself diverges (0.2–0.4 → 0.8–2.1) and every task falls together, *including the one being trained* (T_2: 0.71 → 0.29, 0.69 → 0.30, 0.67 → 0.51 across seeds), before the run recovers; the per-seed depth scales with the divergence, which is where the ±14.5 pp spread comes from. Because all tasks crash together, the depth and area of the three-task group-norm rows record this instability, not the past-task transient the rest of the study measures. Two controls isolate the trigger. The vanilla run crosses the same boundary with a flat replay loss and 4.4 pp of depth, so the near-identical boundary is harmless on its own; the batch-norm run executes the identical schedule through the same boundary, with the same ill-conditioned early ratio, without instability (Figure C.7), so the noise-driven λ is not sufficient either. The failure needs both the unconditioned signal and the group-norm backbone's boundary sensitivity, consistent with the far deeper group-norm transients vanilla ER shows on the same corruption family (shot noise: 46.3 vs 13.1 pp, Table C.8). We therefore read the episode as a reliability boundary of the gradient-norm-ratio signal rather than of the gate principle: the ratio loses its meaning when both norms become small, and the design admits direct guards: freezing λ when both norms fall below a reference scale, or reading a damage signal that does not vanish at a quiet boundary, such as the replay-loss excess over its pre-switch level.

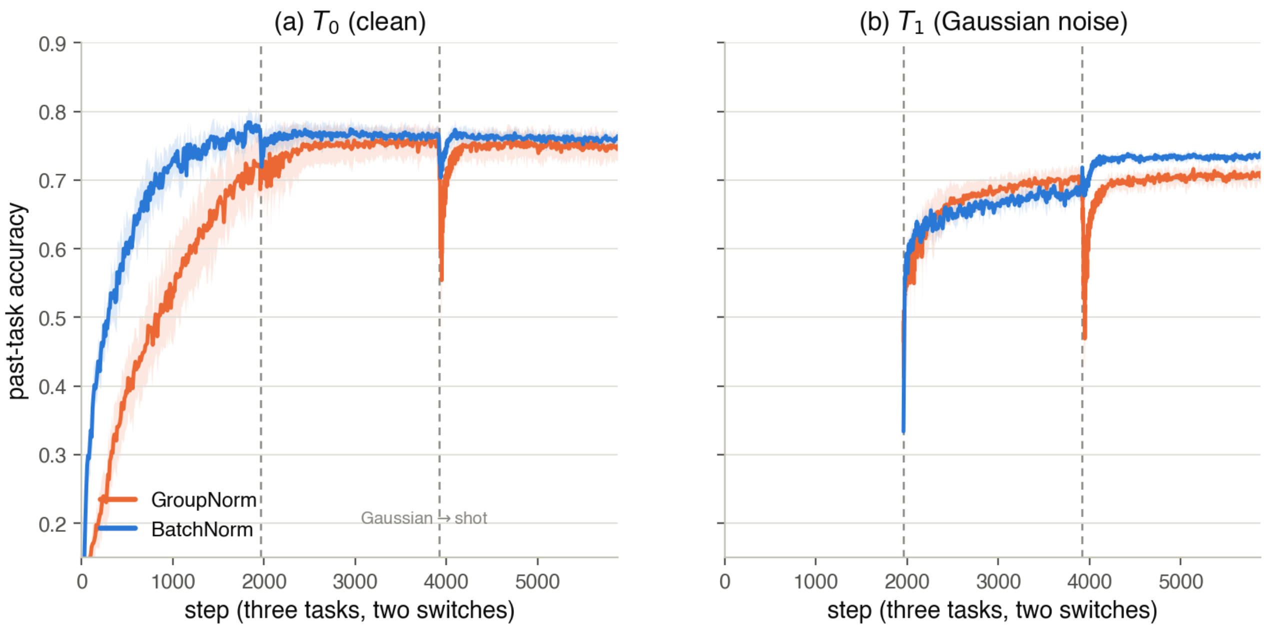


Figure C.7: Per-step past-task accuracy across the three-task CIFAR-10 generalisation sequence (none → Gaussian → shot, ResNet-18, practical curriculum, three seeds with ±std bands), comparing the group-norm run (orange) against its batch-norm counterpart (blue). The two switches (dashed) fall at ∼2k and ∼4k steps. The first boundary (none → Gaussian) barely perturbs the past tasks in either norm. The second (Gaussian → shot, two near-identical corruptions) drives a deep transient in T_0 (a) and T_1 (b) under group norm only, while the batch-norm run holds flat through both switches. The dip is thus localised entirely to the second transition and to group norm, the adaptive-ratio reliability boundary analysed above.

C.8 CIFAR-10: Momentum Cross and Compute Cost

This subsection collects the three deep-backbone checks summarised in Section 6.4: the momentum cross (Table C.9), the generalisation block (Table C.8 above), and the compute cost (Table C.10).

Momentum cross. The cross repeats the composability test of Section 5.2.3 on the ResNet, pairing momentum-0 runs at group norm against the momentum-0.9 rows, and it turns out to be only weakly informative. At momentum 0 the ResNet under-converges the past task in ten epochs (vanilla min-ACC 0.11, ACC 0.65), so every momentum-0 depth sits on a degraded baseline. On that baseline both gates improve with momentum by similar margins (curriculum 28.6 → 5.2 pp, asymmetric PER 22.6 → 5.6 pp), so the cross cannot separate the predicted source-attenuation composition from a general convergence gain, and the prediction that momentum re-inflates the directional gate does not appear; the vanilla null (40.5 → 41.2 pp) is directionally consistent with the account but rests on the same degraded baseline. We therefore draw no mechanism conclusion from this cross in either direction: the momentum signatures rest on the rot-MNIST results, where the MLP isolates them cleanly (Sections 6.2 and 6.3), and a clean deep-backbone test would need a converged momentum-0 baseline (Section 7.4).

Method (GroupNorm)	depth, μ=0	depth, μ=0.9
Vanilla ER	40.5 ± 4.0	41.2 ± 13.5
Curriculum ($\lambda_{\min}$=0.20)	28.6 ± 8.8	5.2 ± 2.1
Asym. PER (δ=0.1)	22.6 ± 9.7	5.6 ± 1.1

Table C.9: CIFAR-10 momentum cross (group norm, buffer 5,000, five seeds), gap depth in pp. All momentum-0 cells sit on an under-converged baseline (vanilla min-ACC 0.11), so the cross is read as uninformative for the momentum signatures rather than as partial confirmation (see text).

Compute cost. Cost is measured with a dedicated benchmark: one full two-task run per gate in its headline configuration (curriculum: adaptive schedule, $\lambda_{\min}$=0.20; asymmetric PER: δ=0.1, 25 warm-started CG iterations; momentum 0.9, group norm and buffer 5,000 on CIFAR), five seeds each on a single NVIDIA RTX

PRO 6000, with per-step stability evaluation disabled so the numbers reflect method compute alone. Wall-clock and CPU time come from `getrusage`; GPU utilisation, power, and memory are sampled at 1 Hz with `nvidia-smi`, and energy is the integrated power. On the ResNet-18 the split is stark (Table C.10). The scalar gate adds one gradient-norm ratio and an EMA update to a standard ER step, and its run is not even compute-bound: the GPU sits half-idle at 49% utilisation and 259 W. The directional gate's 25 conjugate-gradient Fisher-vector products per step, each on the order of a backward pass, saturate the same card (96%, 367 W) for 16× as long, and since energy is power times time the two factors compound to a 24× energy gap (13 vs. 320 Wh per run). Memory is not the price: peak GPU memory differs by 11%. On the rot-MNIST MLP both gates finish in under half a minute at near-idle utilisation (the wall-clock is fixed startup, not method compute), so no meaningful ratio exists there and the cost question is decided entirely on the deep backbone. The ratios are the transferable claim; the absolute numbers are specific to this card, and because ∼15–30 s of each run is startup, the training-only ratio is if anything slightly above 16×.

Dataset	Gate	wall [s]	util [%]	power [W]	energy [Wh]	mem [MB]
CIFAR-10	Curriculum (scalar)	198 ± 7	49	259	13	4,203
(ResNet-18)	Asym. PER (δ=0.1)	$3{,}157 \pm 11$	96	367	320	4,679
rot-MNIST	Curriculum (scalar)	25 ± 1	<1	—	0.4	749
(MLP)	Asym. PER (δ=0.1)	29 ± 1	8	—	0.5	769

Table C.10: Compute cost of one full two-task run (mean over five seeds, ± sd on wall-clock; single RTX PRO 6000; per-step evaluation disabled): wall-clock, mean GPU utilisation and power, integrated GPU energy, and peak GPU memory. On the ResNet-18 the directional gate costs ∼16× the wall-clock and ∼24× the energy of the scalar gate at similar peak memory. The MLP rows are dominated by fixed startup on a near-idle GPU, so their power draw is idle draw (omitted) and no ratio is meaningful.

C.9 Long-Sequence rot-MNIST (L-series)

Condition	ACC	FORG	min-ACC	WF_{100}	WP_{100}	WC-ACC	Depth	Area end
Momentum 0.0								
L1 vanilla ER	0.861 ± 0.004	0.060 ± 0.004	0.670 ± 0.055	0.207 ± 0.053	0.443 ± 0.011	0.722 ± 0.043	21.0 ± 5.8	3.14 ± 0.77
L2 curriculum $\lambda_{\min}$=0.20	0.860 ± 0.002	0.031 ± 0.005	0.828 ± 0.007	0.073 ± 0.005	0.400 ± 0.003	0.840 ± 0.005	4.1 ± 0.8	1.62 ± 0.42
L3 asym. PER δ=0.1	0.870 ± 0.003	0.053 ± 0.006	0.843 ± 0.008	0.038 ± 0.007	0.381 ± 0.003	0.862 ± 0.007	4.7 ± 1.2	0.38 ± 0.24
Momentum 0.9								
L1 vanilla ER	0.889 ± 0.007	0.096 ± 0.008	0.845 ± 0.007	0.065 ± 0.008	0.317 ± 0.006	0.870 ± 0.006	8.9 ± 1.7	1.68 ± 0.69
L2 curriculum $\lambda_{\min}$=0.20	0.897 ± 0.002	0.071 ± 0.003	0.871 ± 0.005	0.041 ± 0.004	0.364 ± 0.005	0.888 ± 0.004	5.4 ± 0.6	0.60 ± 0.14
L3 asym. PER δ=0.1	0.884 ± 0.004	0.104 ± 0.004	0.849 ± 0.006	0.054 ± 0.004	0.325 ± 0.005	0.873 ± 0.005	7.5 ± 1.5	0.80 ± 0.63

Table C.11: Five-task rot-MNIST (rotations $[0, 30, 60, 90, 120]^\circ$, four consecutive transitions, five seeds), the L-series: both gates in their practical configurations against vanilla ER, at both momentum settings. Depth is the worst single-boundary drop and area the end-referenced transient, aggregated as in the two-task blocks. At momentum 0 both gates cut the depth from 21 pp to ≈4 pp; under momentum the curriculum keeps the best cumulative retention while the re-inflated directional gate falls slightly behind vanilla on FORG (0.104 vs. 0.096, all five seeds).

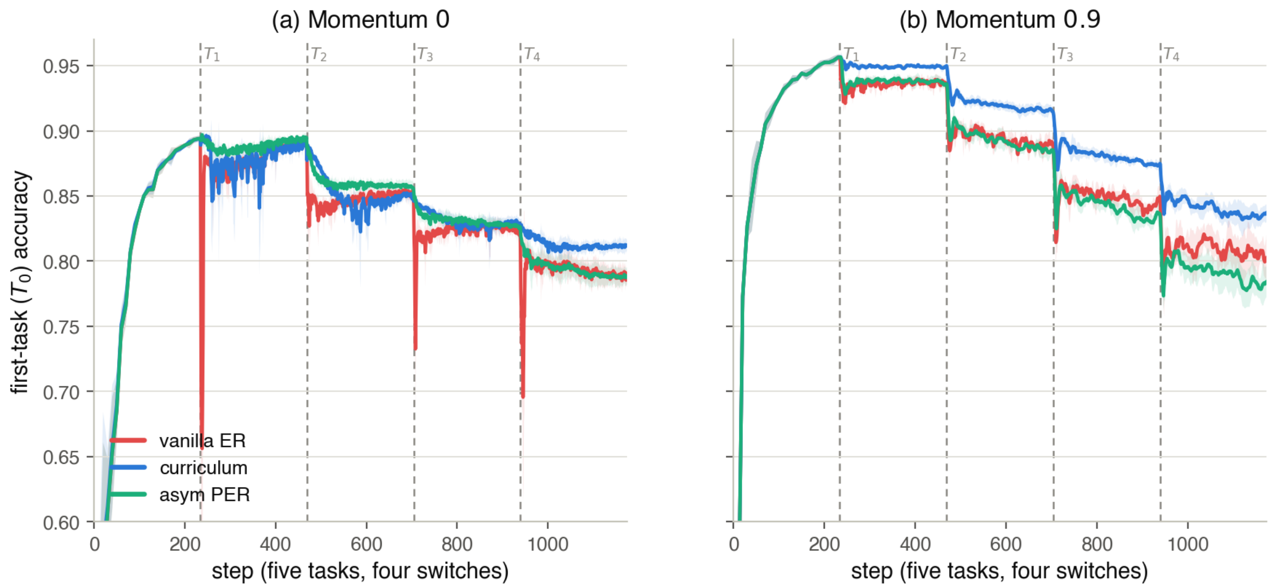


Figure C.8: Five-task rot-MNIST (rotations $[0, 30, 60, 90, 120]^\circ$, four consecutive switches, five seeds with $\pm$std bands), the L-series: first-task T_0 accuracy across the whole sequence for vanilla ER (red), the practical curriculum $\lambda_{\min}$=0.20 (blue) and asymmetric PER δ=0.1 (aqua); dashed lines mark the four switches, and the two panels are the two momentum settings. (a) Momentum 0: both gates hold T_0 almost flat through every boundary, asymmetric PER the smoothest, while vanilla ER drops sharply at each switch. (b) Momentum 0.9: PER's boundary transients return as momentum re-inflates its per-step filter (Section 6.3); the curriculum keeps the boundaries smoothest and holds the highest plateau (Table C.11).

D Derivation of the Trajectory Bound

This appendix derives the two results used in Section 3.1: the replay loss is monotone along the reference path (Proposition 1), and a gradient-flow iterate under a linear ramp tracks that path with $O(1/N)$ lag (Proposition 2), which together give the transient bound of Equation (3.2).

Setup. For $\lambda \in [0, 1]$ define the curriculum objective

$$L_\lambda(\theta) \;=\; L_{\text{replay}}(\theta) + \lambda\, L_{\text{current}}(\theta), \qquad H_\lambda(\theta) \;=\; H_{\text{replay}}(\theta) + \lambda\, H_{\text{current}}(\theta), \tag{D.1}$$

with $H_{\text{replay}} = \nabla^2 L_{\text{replay}}$ and $H_{\text{current}} = \nabla^2 L_{\text{current}}$. We make two assumptions.

(A1) L_{replay} and L_{current} are C^2 in a neighbourhood of the path considered below.

(A2) There is a C^1 branch of critical points $\lambda \mapsto \Theta^*(\lambda)$ of L_λ with $\Theta^*(0) = \theta_0^*$ and $H_\lambda(\Theta^*(\lambda)) \succ 0$ for all $\lambda \in [0, 1]$.

Assumption (A2) asks for a tracked branch of strict local minima, not global convexity. Near $\lambda = 0$ it holds automatically: θ_0^* is a non-degenerate local minimum of L_{replay}, so $H_0(\theta_0^*) \succ 0$ and the implicit function theorem yields a unique C^1 branch locally. Its persistence is controlled by Weyl's inequality, $\mu_{\min}(H_\lambda) \geq \mu_{\min}(H_{\text{replay}}) + \lambda\, \mu_{\min}(H_{\text{current}})$, with $\mu_{\min}$ the smallest eigenvalue: positive-definiteness survives until the new task's negative curvature outweighs the old task's, and holds for all λ whenever $H_{\text{current}} \succeq 0$ along the branch. On a deep non-convex backbone (A2) is a local modelling assumption, which is why Section 3.1 presents the bound as motivation rather than a guarantee.

Proposition 1 (Monotonicity of the replay loss). *Under (A1)–(A2),*

$$\frac{d}{d\lambda}\, L_{replay}(\Theta^*(\lambda)) \;=\; \lambda\, \nabla L_{current}^\top H_\lambda^{-1}\, \nabla L_{current} \;\geq\; 0 \qquad \text{for all } \lambda \in [0, 1],$$

where $\nabla L_{current}$ and H_λ are evaluated at $\Theta^*(\lambda)$. In particular $L_{replay}(\Theta^*(\lambda)) \leq L_{replay}(\theta_{joint}^*)$ for all λ, with $\theta_{joint}^* = \Theta^*(1)$.

Proof. The branch satisfies the first-order condition

$$\nabla L_{\text{replay}}(\Theta^*(\lambda)) + \lambda\, \nabla L_{\text{current}}(\Theta^*(\lambda)) = 0. \tag{D.2}$$

Differentiating (D.2) in λ and collecting terms gives $H_\lambda \frac{d\Theta^*}{d\lambda} + \nabla L_{\text{current}} = 0$, hence

$$\frac{d\Theta^*}{d\lambda} = -\, H_\lambda^{-1}\, \nabla L_{\text{current}}. \tag{D.3}$$

By the chain rule and (D.2),

$$\frac{d}{d\lambda}\, L_{\text{replay}}(\Theta^*(\lambda)) = \nabla L_{\text{replay}}^\top \frac{d\Theta^*}{d\lambda} = (-\lambda\, \nabla L_{\text{current}})^\top \left(-H_\lambda^{-1}\, \nabla L_{\text{current}}\right) = \lambda\, \nabla L_{\text{current}}^\top H_\lambda^{-1}\, \nabla L_{\text{current}},$$

which is non-negative because $\lambda \geq 0$ and $H_\lambda^{-1} \succ 0$ by (A2). Integrating from λ to 1 gives the final statement. □

Proposition 2 ($O(1/N)$ tracking lag)**.** *Let $\theta(t)$ follow the gradient flow $\dot\theta = -\nabla L_{\lambda(t)}(\theta)$ under the linear ramp $\lambda(t) = t/N$, with $\theta(0) = \theta_0^*$, and let $e(t) = \theta(t) - \Theta^*(\lambda(t))$ denote the tracking error. Suppose, in addition to (A1)–(A2), that along the branch $\mu_{\min}(H_\lambda) \geq \mu > 0$ and $\|\nabla L_{current}\| \leq g$. Then, to first order in e, the error settles at*

$$\|e\| \;\leq\; \frac{1}{N}\, \|H_\lambda^{-1}\|\, \left\|\frac{d\Theta^*}{d\lambda}\right\| \;\leq\; \frac{g}{\mu^2 N} \;=\; O\!\left(\frac{1}{N}\right).$$

Proof. Differentiating the error and inserting the flow and the chain rule for the moving target,

$$\dot e = -\nabla L_{\lambda(t)}(\theta(t)) - \frac{d\Theta^*}{d\lambda}\, \dot\lambda, \qquad \dot\lambda = \frac{1}{N}.$$

Linearising the gradient about the minimiser, $\nabla L_\lambda(\theta) = H_\lambda\, e + O(\|e\|^2)$ since $\nabla L_\lambda(\Theta^*) = 0$, yields the error dynamics

$$\dot e \;\approx\; -H_\lambda\, e \;-\; \frac{1}{N}\, \frac{d\Theta^*}{d\lambda}.$$

Because $H_\lambda \succeq \mu I$, these dynamics contract toward the quasi-static equilibrium $e^\dagger = -\frac{1}{N} H_\lambda^{-1} \frac{d\Theta^*}{d\lambda}$; bounding $\|H_\lambda^{-1}\| \leq 1/\mu$ and, by (D.3), $\|d\Theta^*/d\lambda\| \leq g/\mu$ gives the stated bound. The argument is a first-order, continuous-time approximation: it captures how the lag scales with the ramp length N, not an exact discrete-time constant. □

The transient bound. Expanding the replay loss about the reference point, $L_{\text{replay}}(\theta(t)) = L_{\text{replay}}(\Theta^*) + \nabla L_{\text{replay}}(\Theta^*)^\top e + O(\|e\|^2)$, Proposition 1 bounds the first term by $L_{\text{replay}}(\theta^*_{\text{joint}})$ and Proposition 2 bounds the correction by $O(1/N)$ (the gradient norm $\|\nabla L_{\text{replay}}(\Theta^*)\| = \lambda \|\nabla L_{\text{current}}\| \leq g$ is bounded along the branch). Hence

$$\max_t\, L_{\text{replay}}(\theta(t)) \;\leq\; L_{\text{replay}}(\theta^*_{\text{joint}}) + O\!\left(\frac{1}{N}\right) : \tag{D.4}$$

the permanent price is the first term, and the transient contribution of the tracking lag shrinks as the ramp lengthens, which is Equation (3.2).